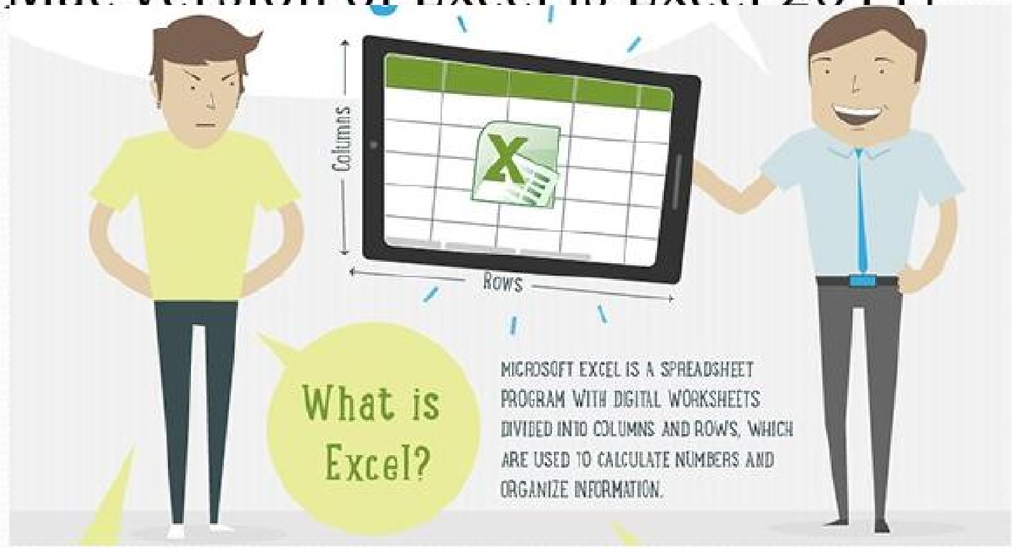
Excel Basics

RV Misa

What is Excel?

- Excel is a spreadsheet application software from Microsoft
- Some spreadsheet programs are also available free of charge. For example, Google Docs application suite
- Beginning with Excel 2 for Windows, many versions of Excel have appeared so far. Excel 2013 is the latest version for Windows. [The latest Mac version of Excel is Excel 2011]



Spreadsheet Origin

- Dan Bricklin and Bob Frankston
  - Invented in 1979
  - VisiCalc for Apple II
    - Took 20 hours of work per week from some people and made it into 15 minutes of work
    - Sold and developed into Lotus 1-2-3

Excel 2007 & Beyond

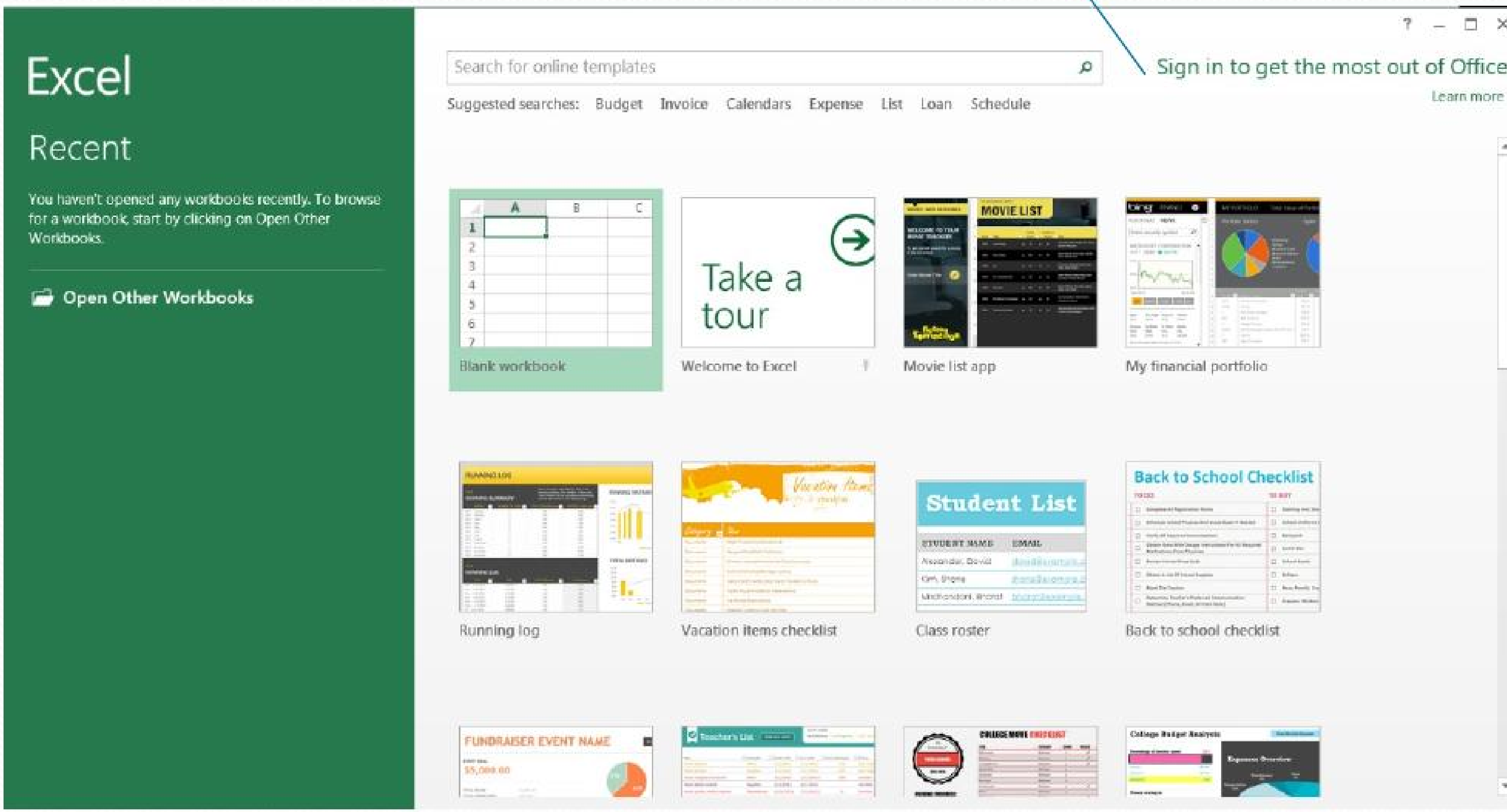
- Excel 2007 and versions thereafter differ significantly from earlier versions of Excel
  - These versions offer a totally different look and feel of the user interface from earlier versions
  - Expanded features and capabilities with every new release

Getting Familiar with Excel

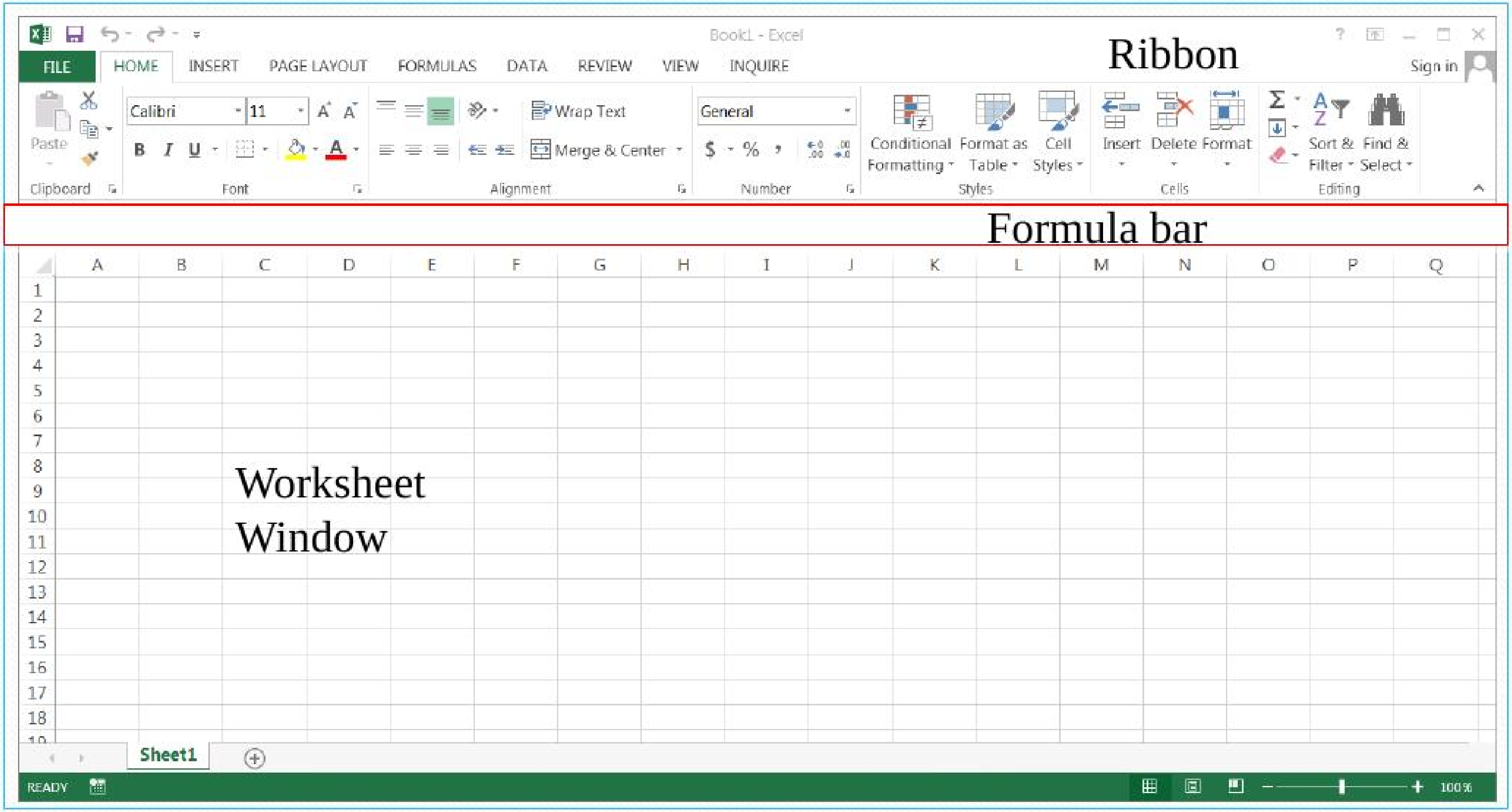
- Start screen
- Excel interface elements
- Backstage view
- Workbooks and worksheets
- Moving around in a worksheet
- Data in Excel
- Excel Formulas
- Formatting

Start Screen

You can save your files in **OneDrive**, a built-in cloud support in Office 2013



Excel Interface Elements



The Ribbon

Common across all applications in Microsoft Office for consistent look and feel.

File button

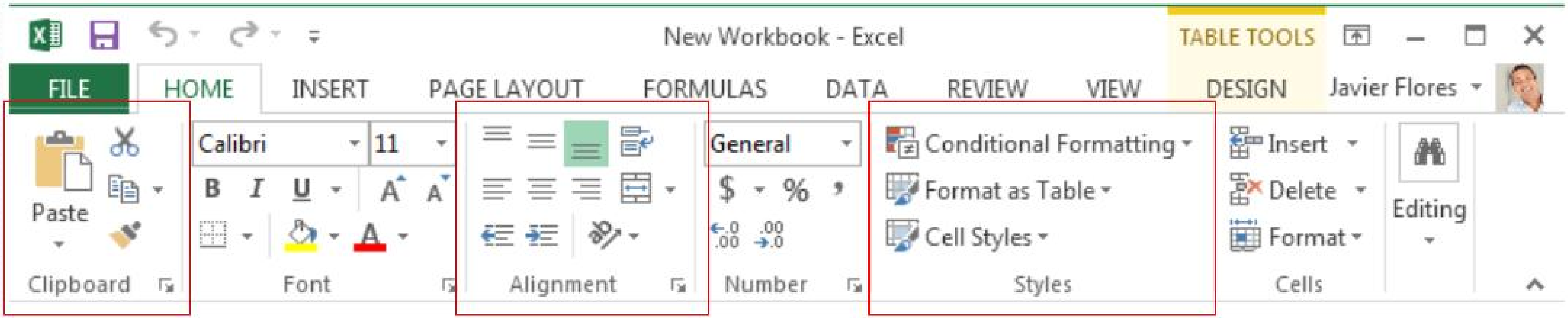
Tabs

Only one tab is active at any time. The active tab is highlighted

Tabs

- Only one tab can be active. The active tab is shown highlighted. You can make a tab active by clicking on it.
- The Ribbon shows a set of panels below the tabs row
- The set of panels shown corresponds to the active tab. If you make another tab active, the panel set changes
- Each panel shows a group of related buttons or icons
- Some tabs appear only based upon certain actions. Such tabs are known as contextual tabs

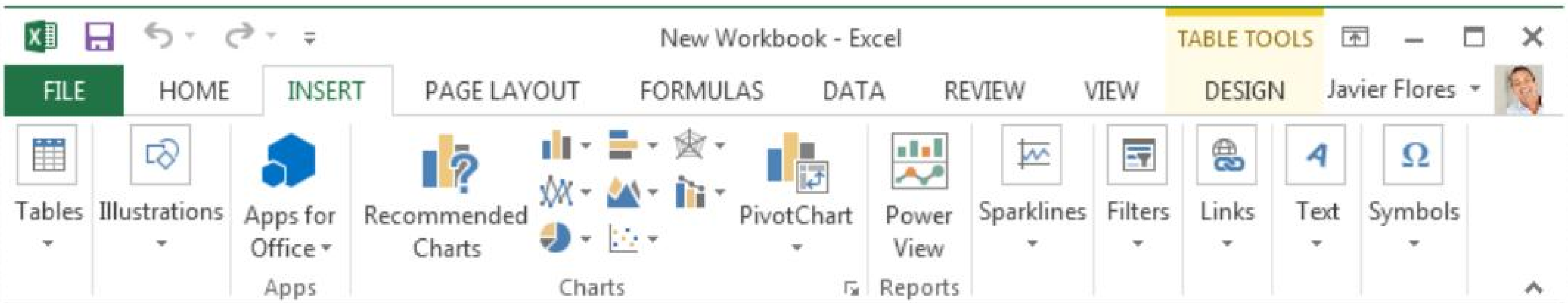
The Ribbon



The Home tab gives you access to some of the most commonly used commands for working with data in Excel 2013, including copy and paste, formatting, and number styles. The Home tab is selected by default whenever you open Excel.

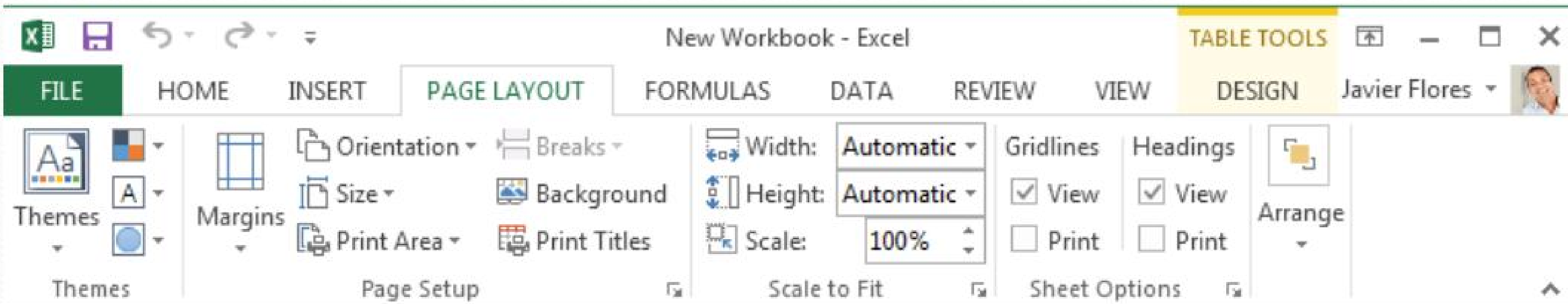
Panels. Each panel has buttons for related commands

Insert Tab



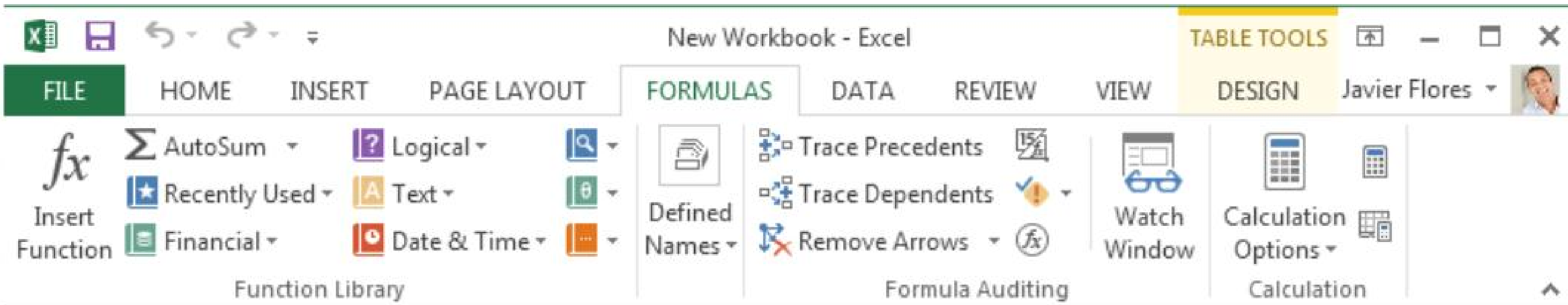
The **Insert** tab allows you to insert **charts, tables, sparklines, filters,** and more, which can help you **visualize** and **communicate** your workbook data **graphically.**

Page Layout Tab



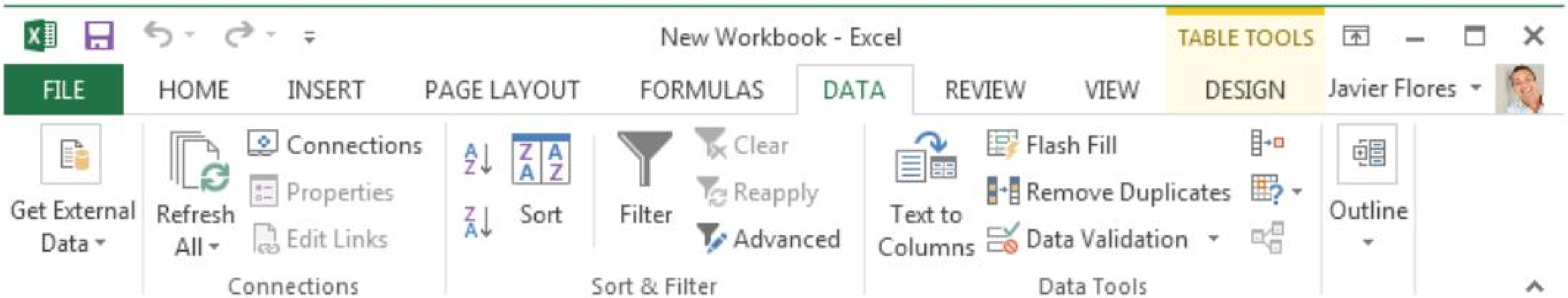
The **Page Layout** tab allows you to change the **print formatting** of your workbook, including **margin width, page orientation,** and **themes.** These commands will be especially helpful when preparing to print a workbook.

Formulas Tab



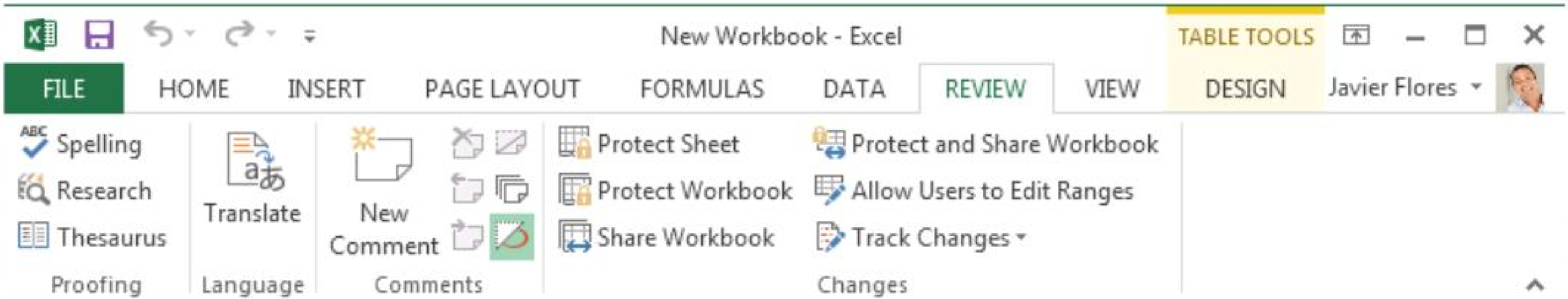
The **Formulas** tab gives you access to the most commonly used **functions** and **formulas** in Excel. These commands will help you **calculate** and **analyze numerical data,** like averages and percentages.

Data Tab



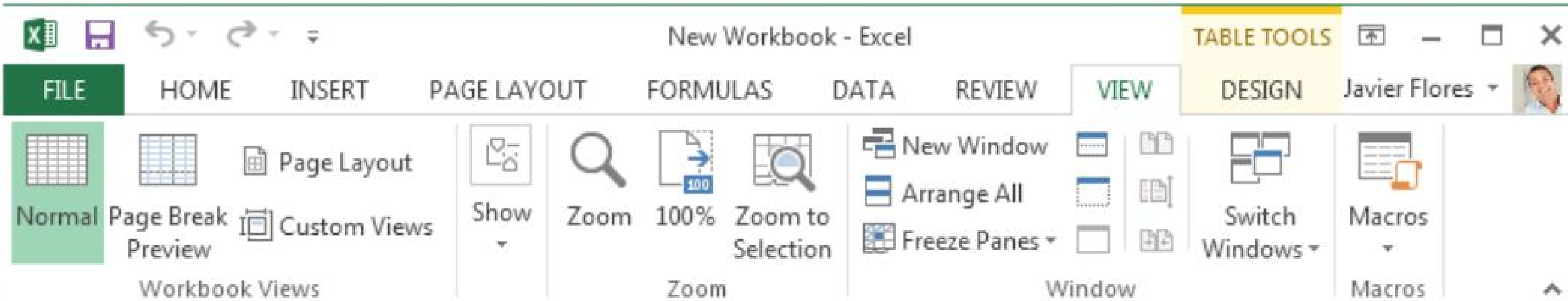
The Data tab makes it easy to sort and filter information in your workbook, which can be especially helpful if your project contains a large amount of data.

Review Tab



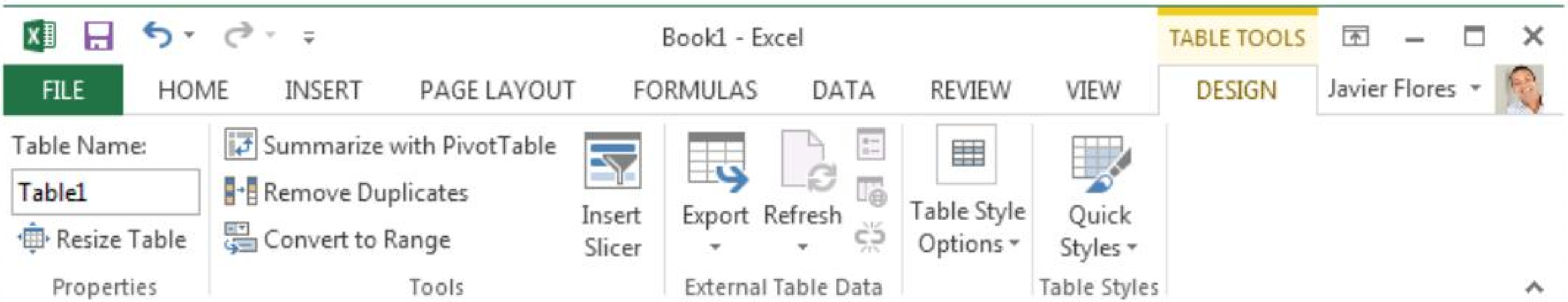
You can use the **Review** tab to access Excel's powerful **editing features,** including **comments** and track **changes.** These features make it easy to **share** and **collaborate** on workbooks.

View Tab



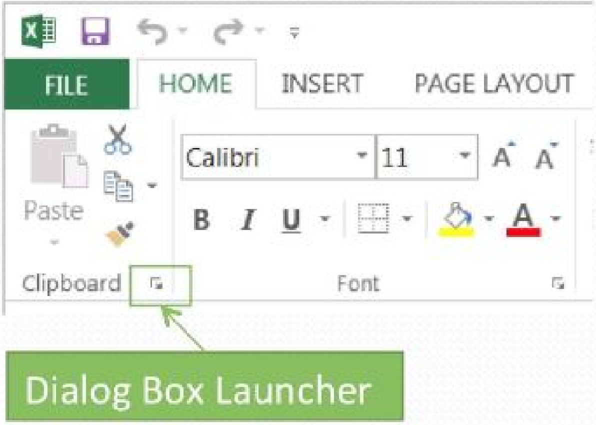
The **View** tab allows you to switch between different **views** for your workbook and **freeze panes** for easy viewing. These commands will also be helpful when preparing to print a workbook.

Contextual Tab

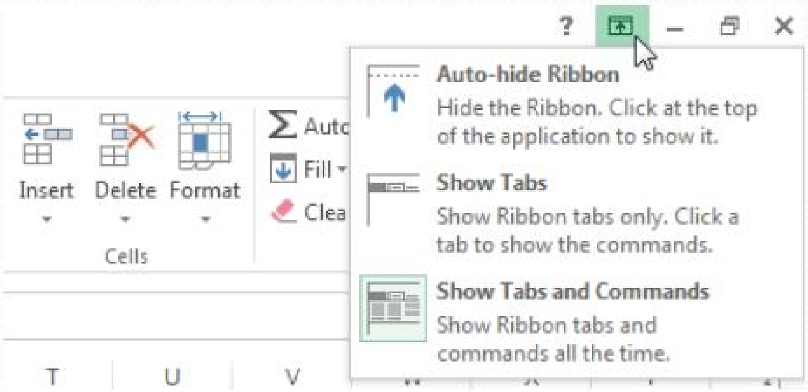


Contextual tabs will appear on the Ribbon when working with certain items, like **tables** and **pictures**. These tabs contain special command groups that can help you format these items as needed.

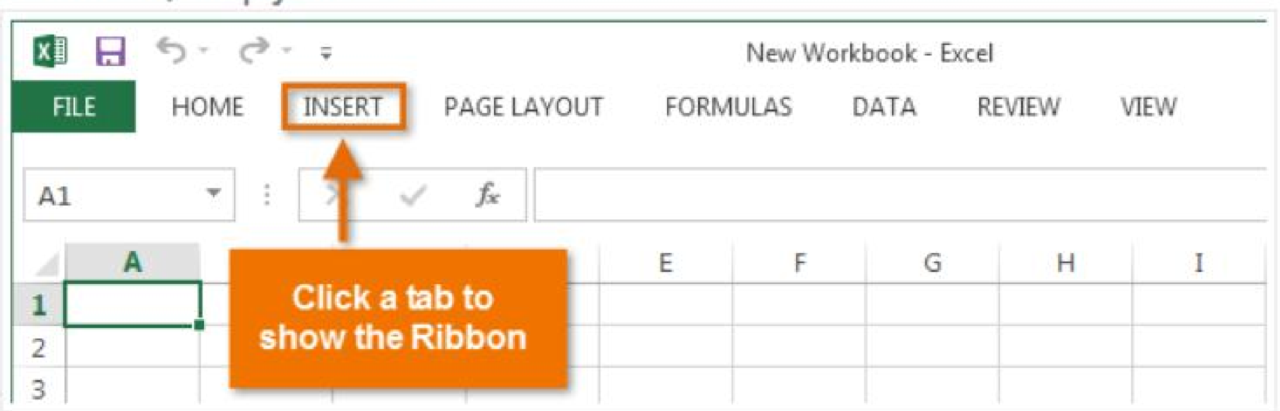
Dialog Box Launcher in Panels



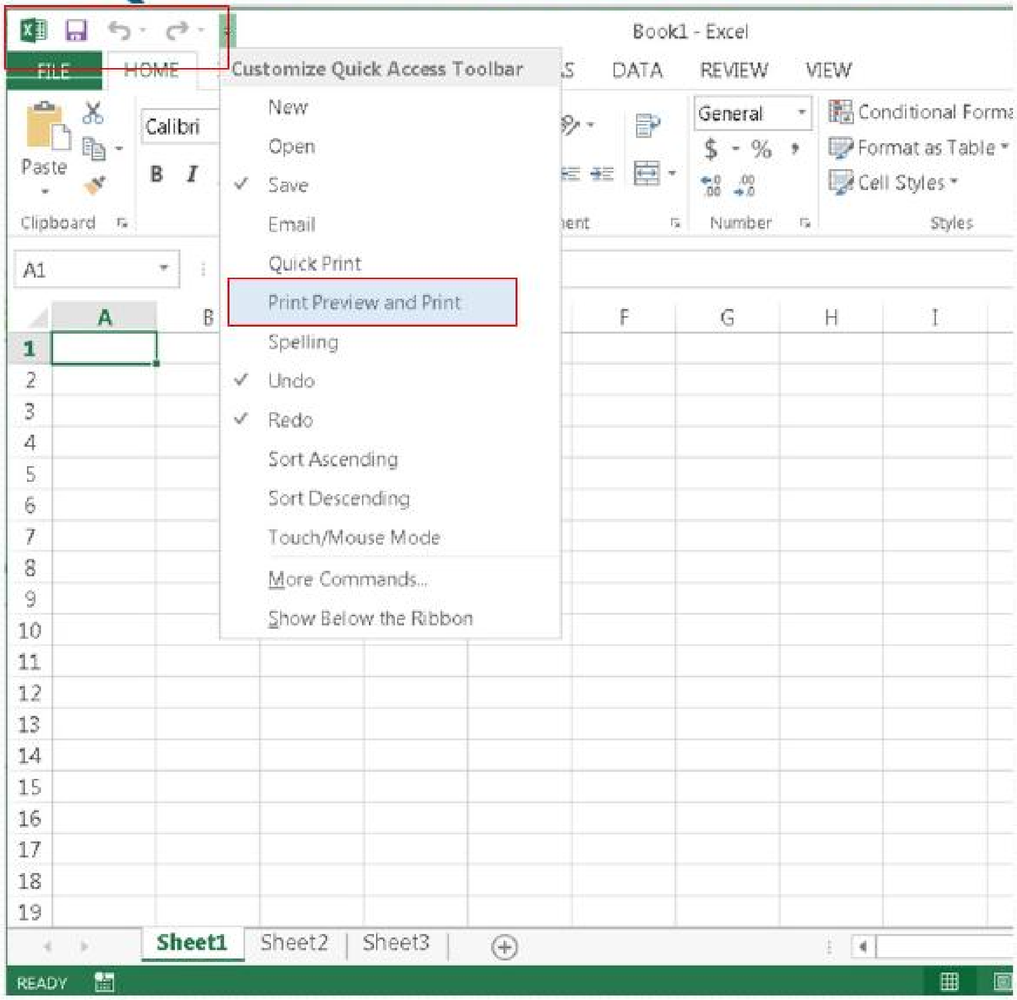
The Ribbon Display Options Button

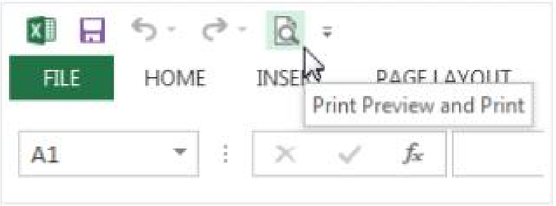


Show Tabs: This option hides all command groups when not in use, but **tabs** will remain visible. To **show the Ribbon**, simply click a tab.

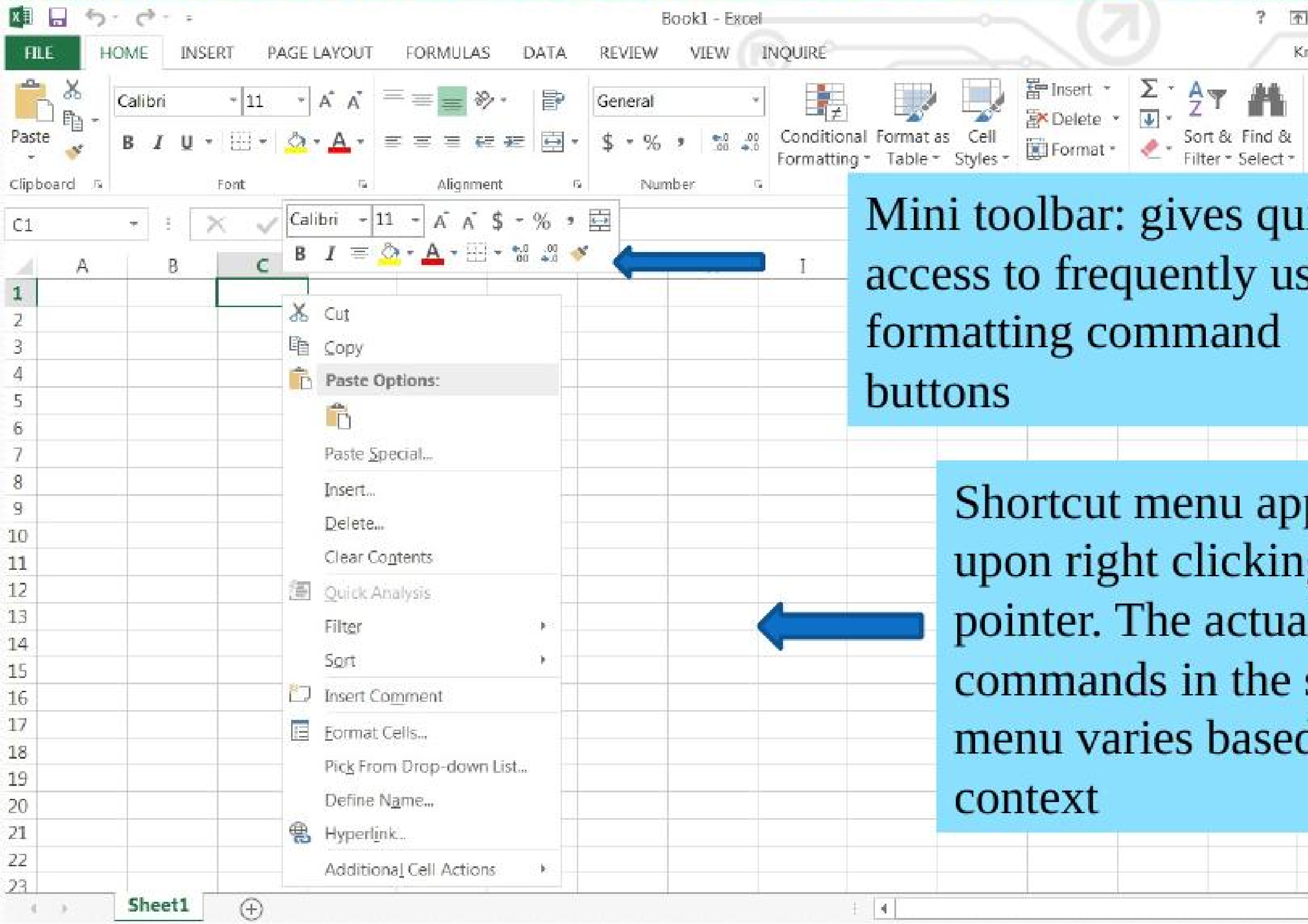


Quick Access Toolbar (QAT)





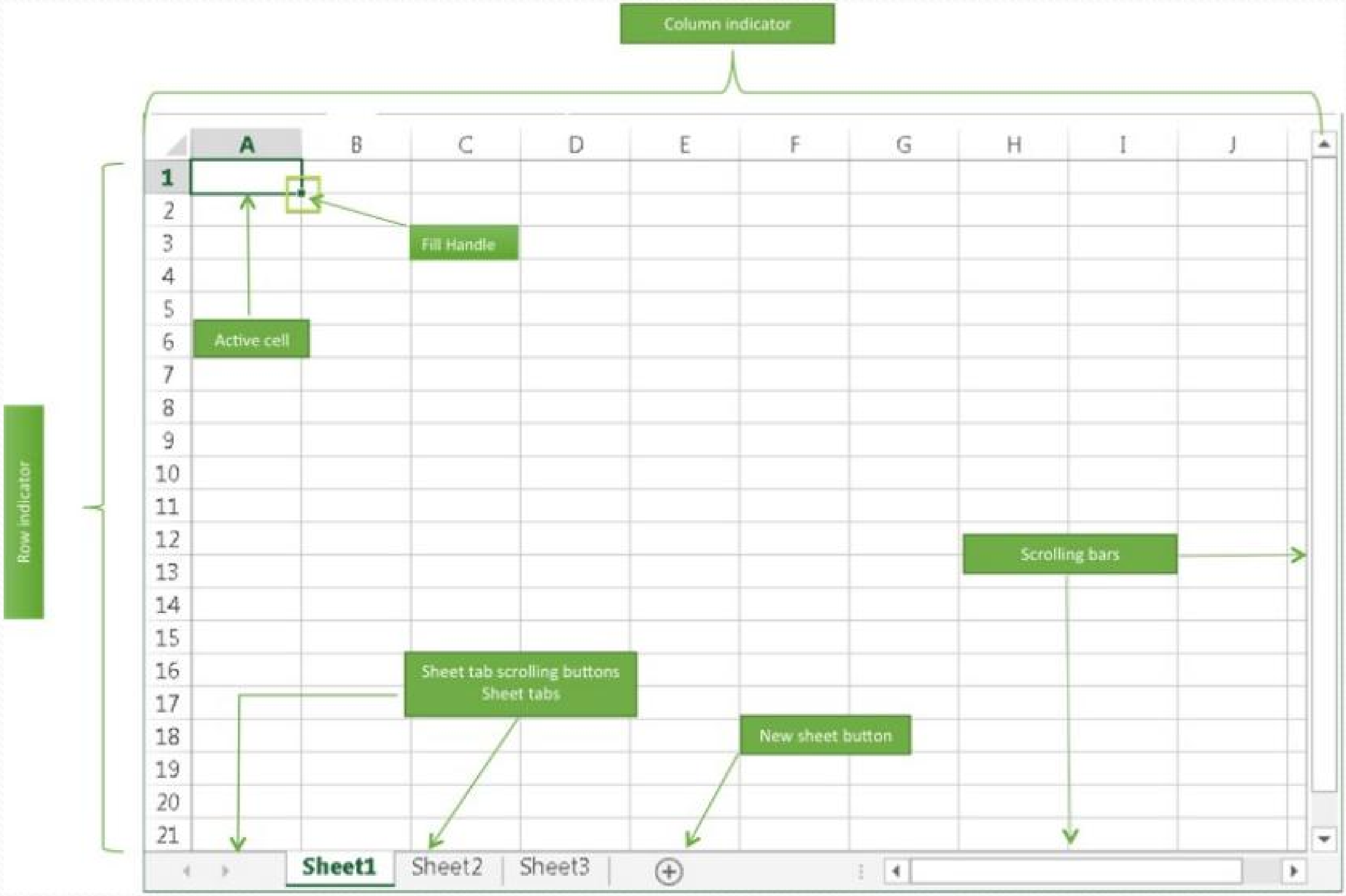
Mini Toolbar & Shortcut Menu Bar



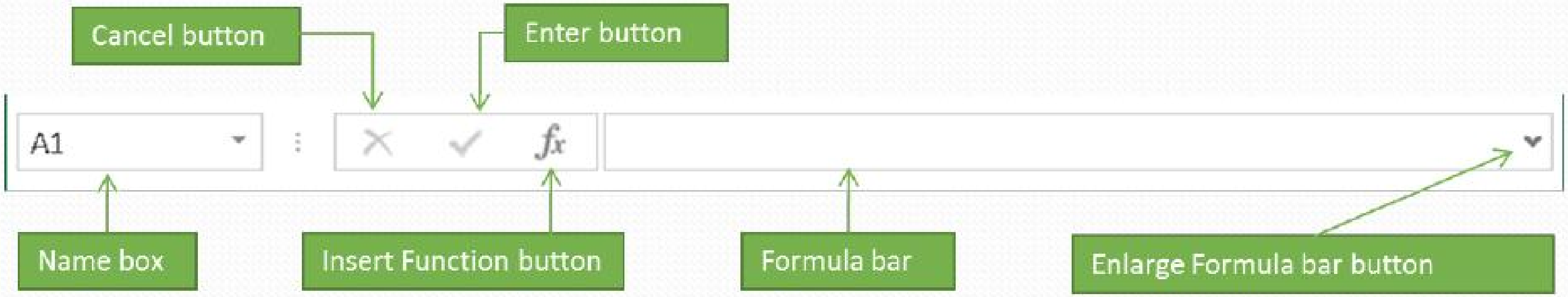
Mini toolbar: gives quick access to frequently used formatting command buttons

Shortcut menu appears upon right clicking the pointer. The actual list of commands in the shortcut menu varies based on context

Worksheet Window

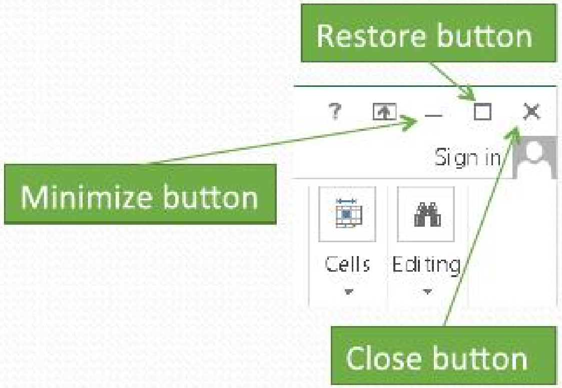


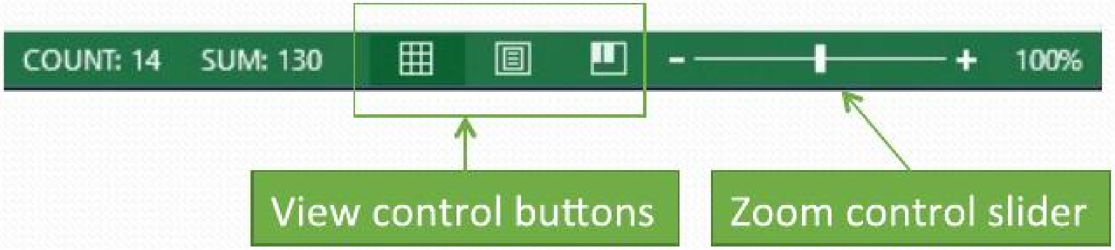
Formula Bar



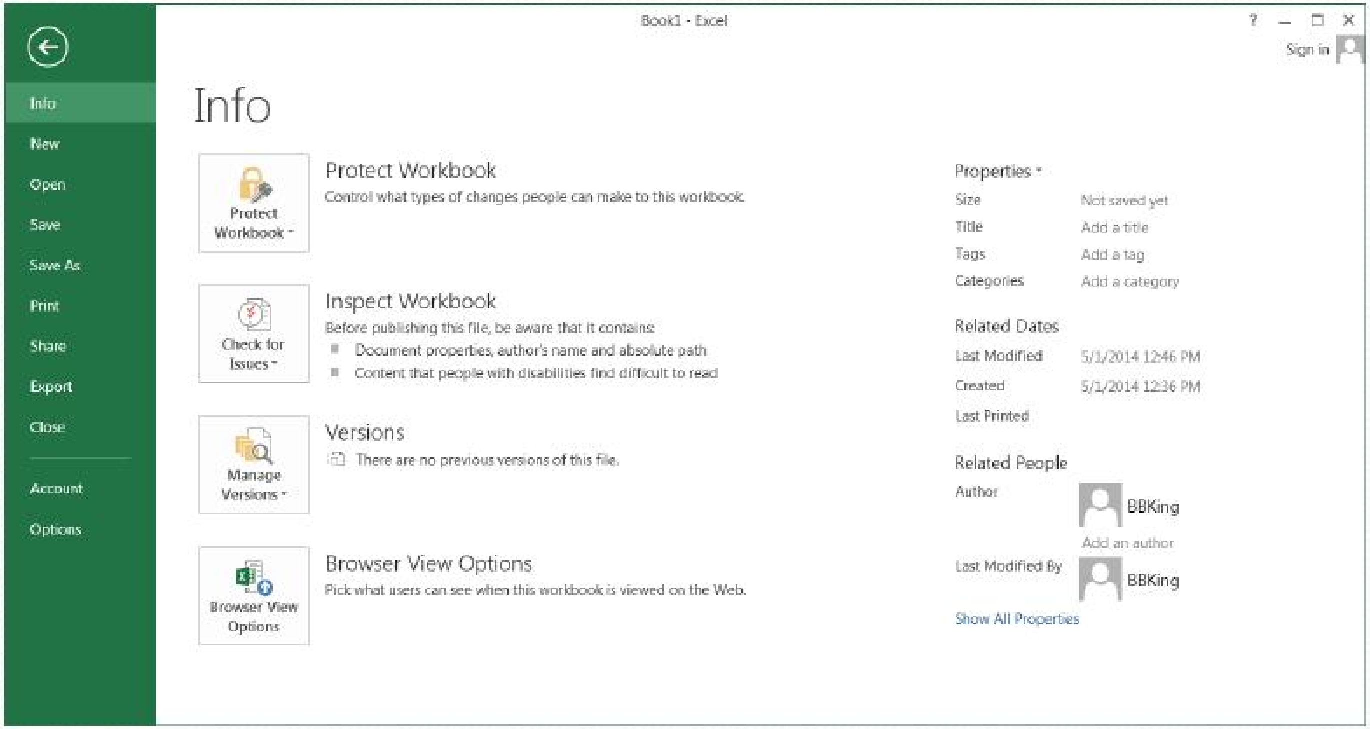
Cancel button
Enter button
A1
fx
Name box
Insert Function button
Formula bar
Enlarge Formula bar button

Control Buttons and Status Bar





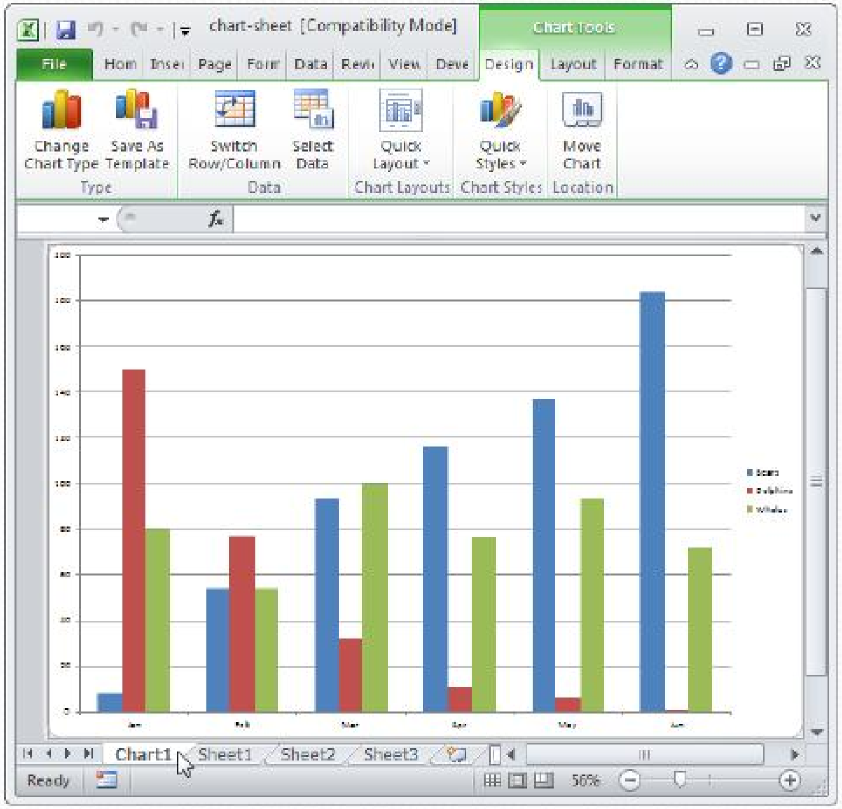
Backstage View



Backstage view appears when you click the File tab.

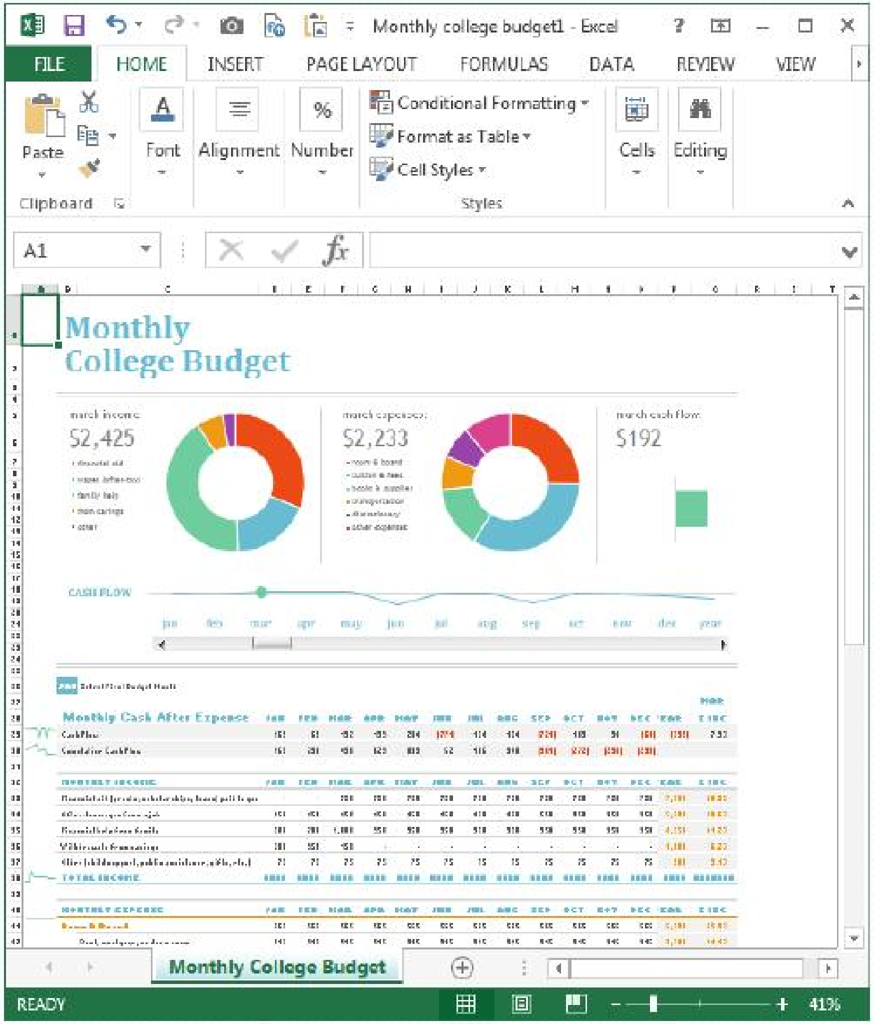
Workbooks and Worksheets

- An Excel workbook is made up of worksheets and chart sheets. The chart sheets are special sheets for storing charts
- The number of worksheets that a workbook can hold is limited only by the computer memory. By default, Excel 2013 opens a new workbook with only one worksheet with the default name *Sheet1*. (Previous versions had three default worksheets)



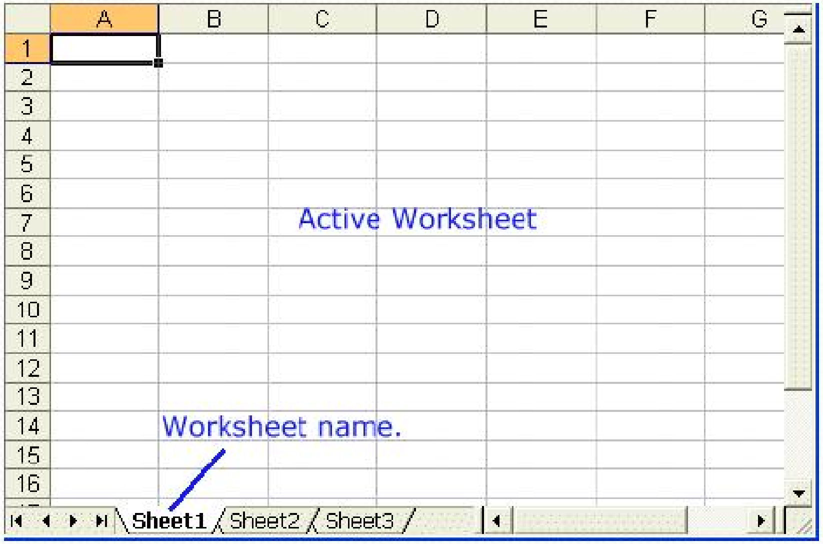
Workbook Templates

- Preformatted workbooks for various tasks with partial content including predefined formulas



Workbooks and Worksheets

- Although many workbooks can be open at any time, only one workbook is designated as the active workbook. Similarly only one worksheet can be the active worksheet at any time
- When you open a new Excel file, it is given the default name of *Book1*. If you open another new file, it will open with name as *Book2*.



Worksheet Specs

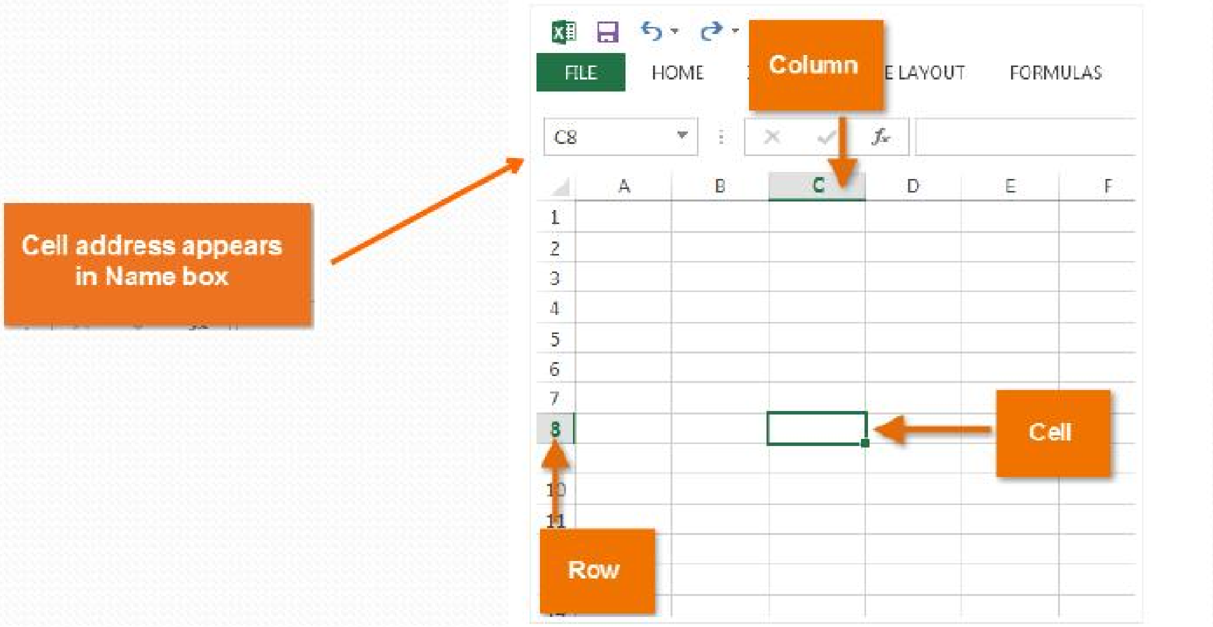
- Every Excel worksheet has 16,384 columns and 1,048,576 rows
- The intersection of a row and a column is called a cell
- The columns are numbered from A to XFD, and rows from 1 to 1,048,576

Mouse Pointer Appearance

- Mouse pointer changes its appearance to indicate what action can be performed
  - Arrow: select item from the Ribbon or scrolling or other commands
  - I-beam: type text in formula bar
  - White plus sign: as the pointer moves over worksheet surface
  - Small black arrow: when the pointer is over the column or row indicators to select a column or a row
  - A cross with double arrow: when placed at the boundary of a selected column or row to change column width or row height

Cells

- Cell reference
  - A cell is referred by the letter for the column in which the cell is located followed by the number of the row holding the cell. Thus, B5 refers to the cell in Column B located in the fifth row. AZ23 means a cell in column AZ and row 23. Cell reference is also known as cell address.

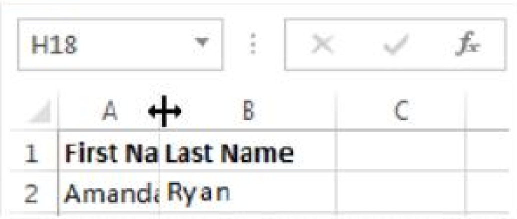


Cells

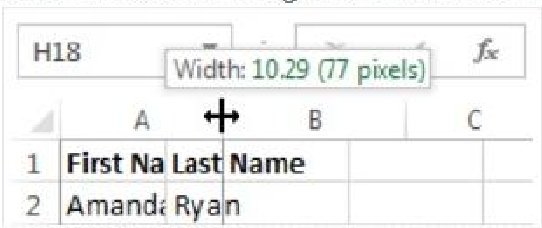
- A cell must be active if we wish to enter data into it. An active cell has a thick dark boundary, called cell selector, along it.
- The bottom right corner of the cell selector is marked by a small square, called the fill handle

Changing Cell Size

1. Position the mouse over the **column line** in the **column heading** so the **white cross** becomes a **double arrow**.

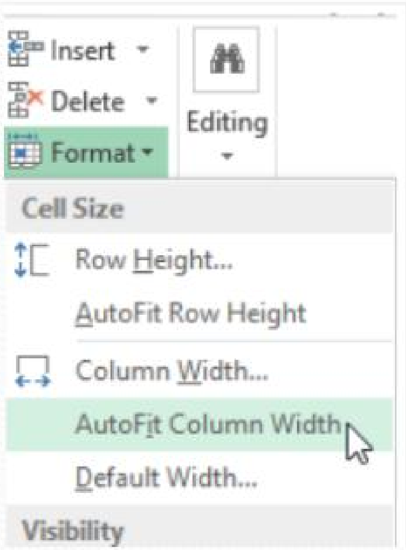


2. Click, hold, and drag the mouse to **increase** or **decrease** the column width.



Changing Cell Size

You can also AutoFit the width for several columns at the same time. Simply select the columns you would like to AutoFit, then select the **AutoFit Column Width** command from the **Format** drop-down menu on the **Home** tab. This method can also be used for **Row height**.



Changing Cell Size for all Cells

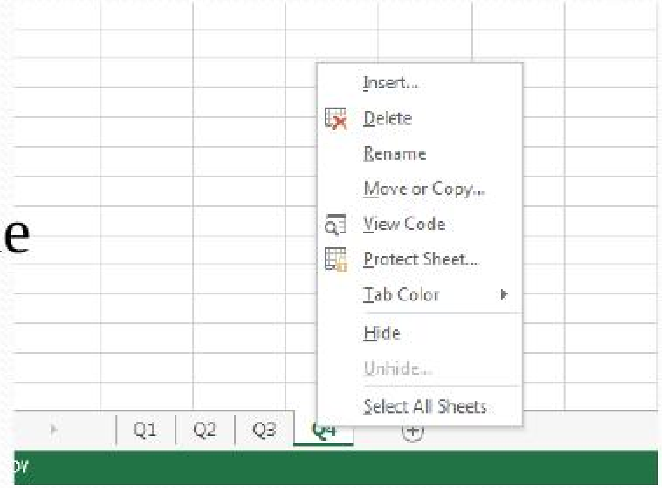
1. Locate and click the **Select All** button just below the **formula bar** to select every cell in the worksheet.

2. Position the mouse over a **row line** so the **white cross** becomes a **double arrow**.
3. Click, hold, and drag the mouse to **increase** or **decrease** the row height.

Renaming, Inserting, and Deleting Worksheets

Excel permits worksheet names limited to 31 characters. Blank spaces are permitted in worksheet names.

To rename a sheet, double-click the sheet tab and enter the new name.

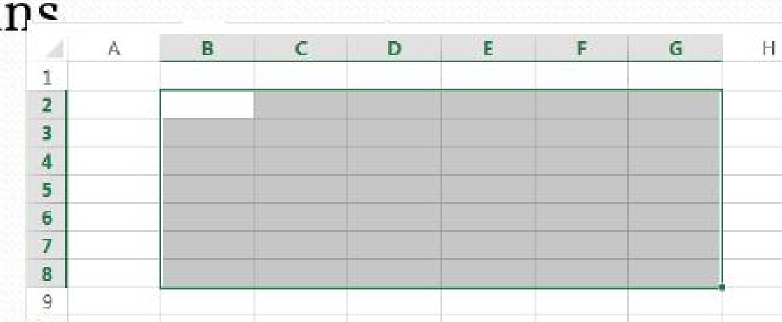


The shortcut menu of commands upon right-clicking a sheet tab

Moving Around a Worksheet

Selecting a Group of Cells

- You can select a group of cells by selecting the top left cell of the group and then dragging the pointer over the cells you want to select
- To select a full row, click on the row number. Do the same to select a column
- To select multiple rows, select the first row and then drag the pointer over row numbers to the desired row. Similar action for multiple columns



A group of selected cells. The first cell in the group in white is the active cell.

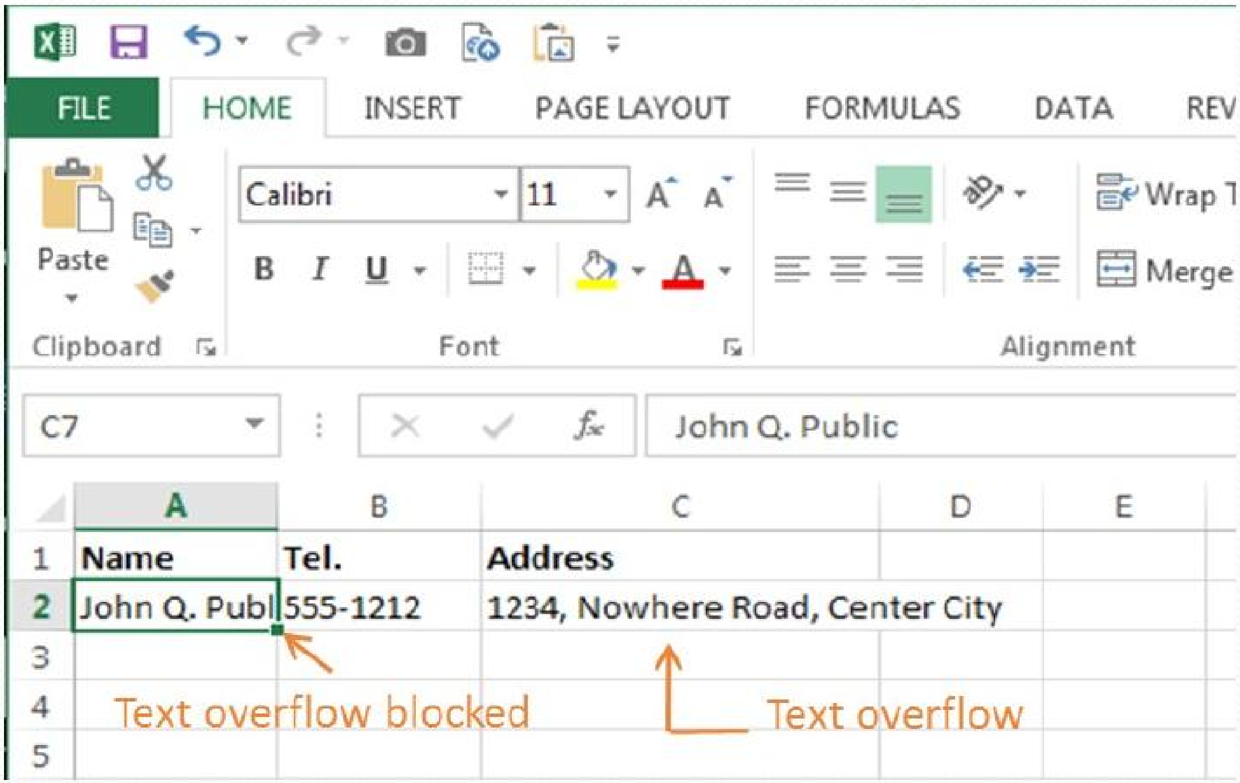
Excel Data Types

- Data types implies the types of cell entries Excel recognizes
- Three different types of entries
  - Text or label
  - Value
  - Formula

Text

- Any combination of letters, numbers, and special characters
- Cannot be used for calculations
- Left aligned in cell (default setting)
- Examples:
  - Names of places/persons
  - Telephone number
  - Security number
  - Column headings, for example Monthly sales

Text Within a Cell



Value Entries

- Numbers, dates, times
- Can be used for calculations
- Right justified in cell (default setting)
- Examples:

Number	Date	Time	Negative Number
378	11/29/94	4:40:31	(9876)

3/15/08 is a recognized as a valid date and hence a valid value entry

15/15/08 is treated as a text entry because 15/15/08 is not a valid date

Value Entries

- Suppose you have an order number 10-16-70. Excel will incorrectly treat it as a valid date (October 16, 1970). In such cases you should enter ‘10-16-70 to let Excel know that it is not a date

Formulas

- A cell entry beginning with an equal sign (=) is treated as a formula in Excel
- A formula is an expression telling Excel to perform an operation
- Excel allows many types of operations; however, we shall consider only arithmetic operations for now
- Examples

=159*3.7

=A1+A2+A3

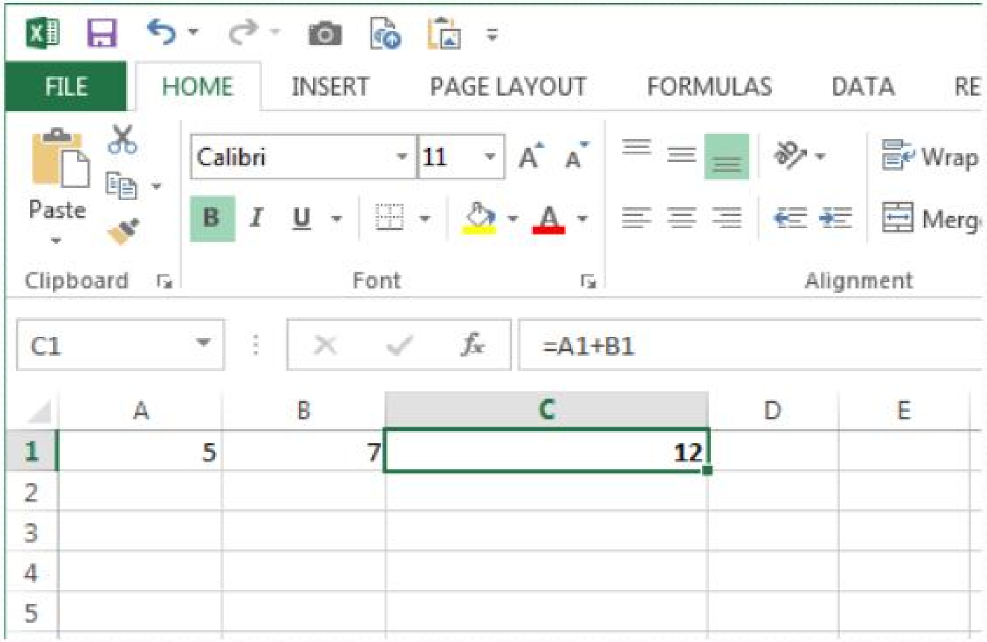
=(2*A1-B1)*C1

=A1/B1+C1^2.5

Operator

Operands

Formula Example



The active cell C1 shows the result; the formula appears in the formula bar

Arithmetic Operators

Parentheses	( )	(5+3)/2	4
Exponentiation	^	5^2	25
Multiplication	*	5*2	10
Division	/	5/2	2.5
Addition	+	5+2	7
Subtraction	-	5-2	3

Formula Examples Showing Operator's Precedence

Formula Example	Calculation Description
=16*3+5	First multiplies 16 and 3, and then adds 5 to produce the result of 53
=A1^3/2	Raises the value in cell A1 to the 3rd power, and then divides the result by 2. For example, if the value in cell A1 is 4, the formula result will be 32
=A1*B1/C1	In this formula, the multiplication between the values of cells A1 and B1 will be carried out first. The multiplication result will be then divided by the value in C1. For example, if the value in cell A1 is 4, the value in B1 is 2, and the value in C1 is 5, then the formula result will be 1.6
=(2*A1-B1)/C1	First multiply 2 with the value in cell A1 and then subtract the value of B1 from it. Next, divide the result by the value in cell C1. Assuming the same set of values for A1, B1, and C1 as above, then the formula result will be 1.2
=A1/B1+C1^2	First the value of cell C1 will be raised to the power 2 to get the first intermediate result. Next the value of cell A1 will be divided by the value in cell B1 to get the second intermediate results. Finally, the two intermediate results will be added to get the final result. Assuming the same set of values for A1, B1, and C1 as above, the formula result will be 27

Parentheses Nesting

- Nesting allows you to tell Excel how a formula should be evaluated. For example in the following formula, the expression within blue parentheses will be evaluated first followed by green and red parentheses

$$=\boxed{(B2*\boxed{(D2\hat{}\boxed{(C2-2)}+A2/C2)}+6.75)}*B4$$

Worksheet Functions

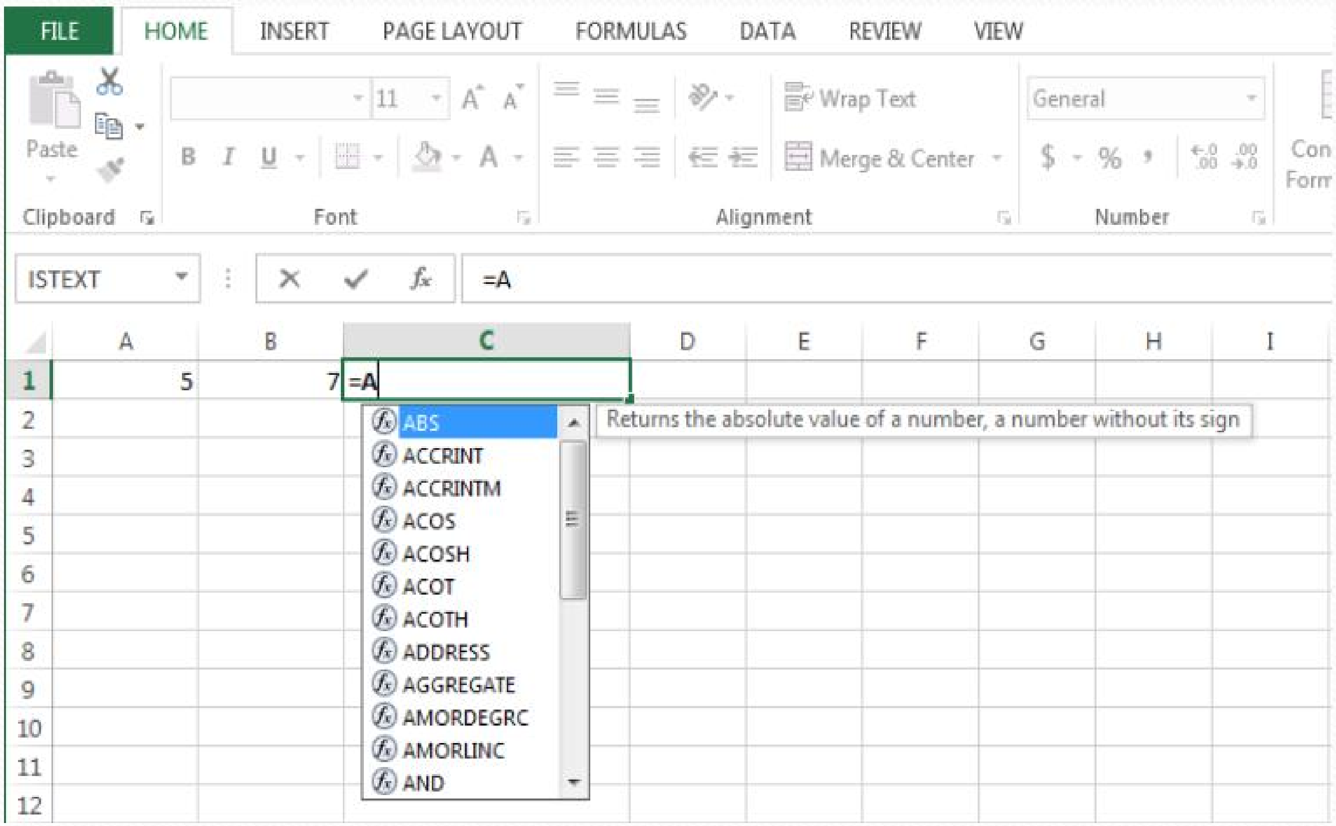
- Excel provides a large number of worksheet functions or simply called functions. We will look at them later.
- Some examples of formulas with functions are:

=SQRT(A1)+5

=SUM(A1,B1,C1)

=SUM(A1,B1,C1)/(SQRT(A1)+5)

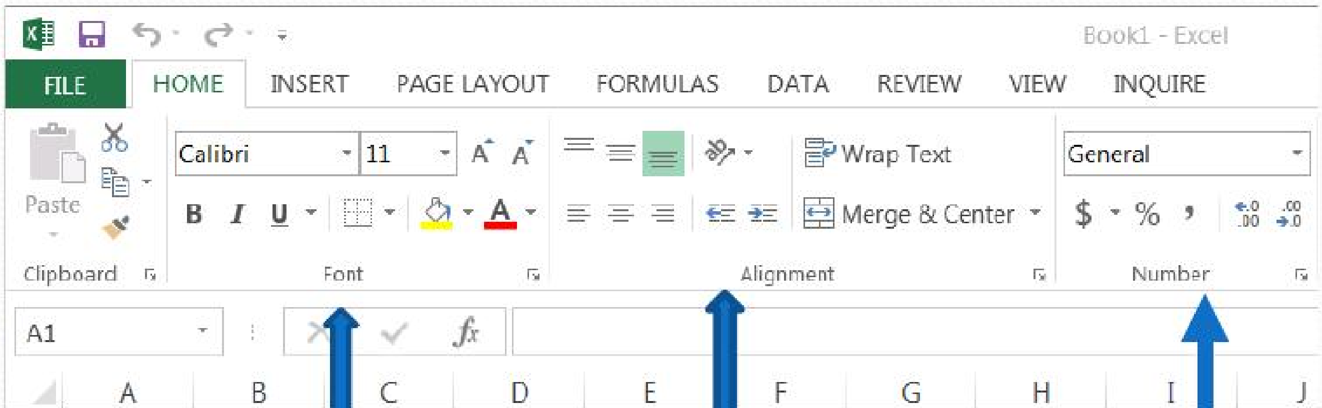
AutoComplete Feature



Formatting

- Controls how information in cells is displayed
- Two aspects of formatting
  - Stylistic formatting
    - Governs font type, size, color, cell background and border style etc
  - Numeric formatting
    - Governs how a value appears in a cell. For example, the number of digits after a decimal point

Formatting Commands Panels



Font Panel

Alignment Panel

Number Panel

Cell Styles Command

This command allows several formatting options, all at once.

Creating a Worksheet

- We want to create a worksheet that:
  - Shows name, id, and the semester of a student at a university
  - Shows the courses taken by the student
  - Shows the credit hours and the grades obtained
  - Calculates the grade point average (GPA)
- The final worksheet should look similar to as shown in the next slide

E15

	A	B	C	D	E	F
1–4	University of Oakmark at Sunny Hills					
5	Name: John Q. Public				Winter Semester	
6	ID# 999999123					
7	CRN Number	Course Title	Credits	Grades	Points	
8						
9	CSE 120	Intro to Computing	4	3.2		
10	CHM 151	Intro to Chemistry	4	2.7		
11	HST 130	American History	3	3.4		
12	ENG 161	Modern Literature	3	3.1		
13	MUS 122	Intro to Western Music	3	3.5		
14						
15		Total Credit Hours		GPA		
16						

Gradesheet Sheet2 Sheet3

Ready 100%

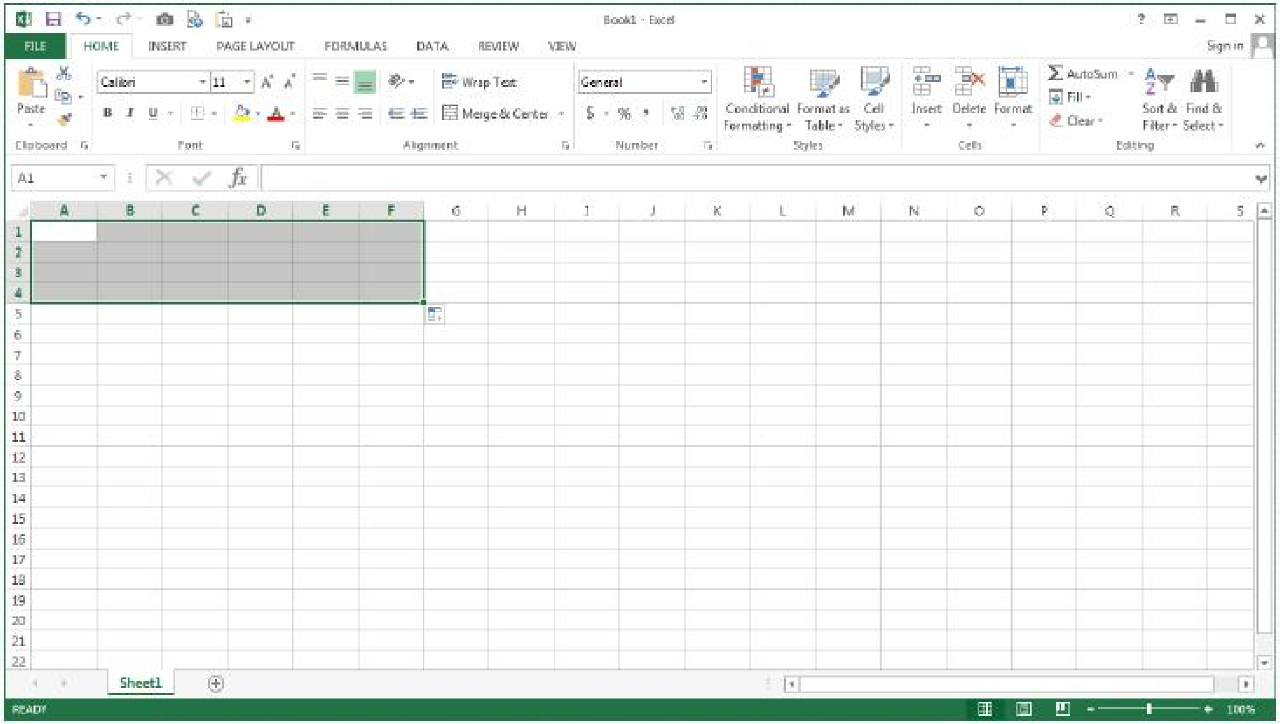
The values in these cells will be calculated by Excel formulas

Step 1: Starting Excel

- Start Excel. You can start Excel by clicking on the Excel icon on your desktop. Alternately, click on the Start button at the bottom left of your Windows desktop, and then point to All Programs to display the programs your computer has. Next, point to Microsoft Office and click on Excel to start it.
- Excel will open a new workbook with the default name *Book1* and cell A1 as the active cell.

Step 2: Formatting Cells A1 to F4

- Select cell A1, click the left button on the mouse and drag it over cells in columns A-F and row 1-4. Your worksheet will appear as shown below



Step 3: Merge Cells A1 to F4

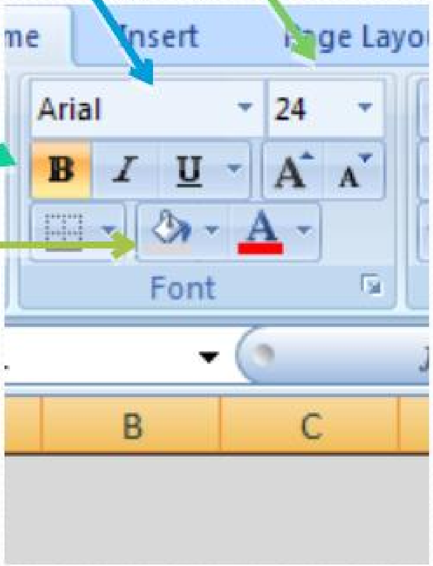
- We will be entering the university name in cells A1 to F4. So we need to merge these cells to act as one large cell
- Click the Merge & Center button in the Alignment panel

Step 4: Set Wrap Text & Alignment

- Click the Wrap Text button to ensure that any text entered in the merged cells will be wrapped around
- Click the Center button in the Alignment panel to instruct Excel that you want text horizontally centered
- Click the Middle Align button to vertically center the text as well

Step 5: Setting Font and Fill Color

- Click on the Font Selection button in the Font panel and select Arial font.
- Set Font size to 24 via the Font Size button
- Select Bold as the font style
- Select a background of your liking by clicking the Fill Color button



Step 6: Enter Information

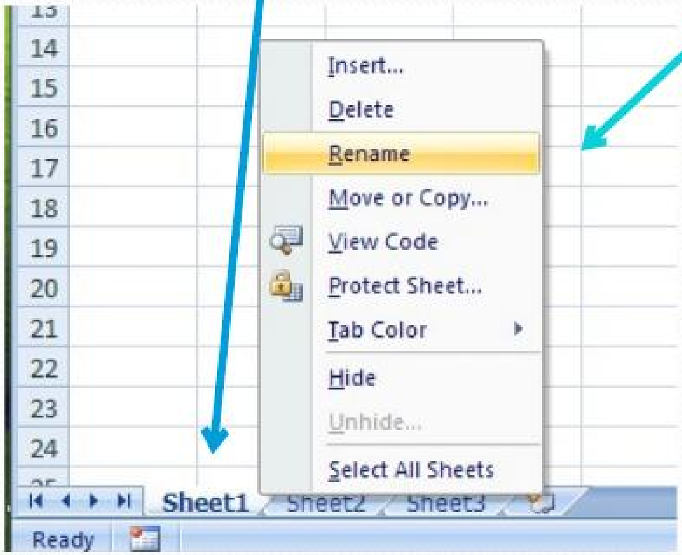
- Enter a name for the university in the merged cells
- Enter the student name in cell A5. Enter the ID and semester information in cells A6 and E5
- Enter the headings in cells A7 to E7
- Enter course numbers, titles, credits, and grades in cells A9 to D13. You can makeup your own courses, credits, and grades, if you desire. You might need to increase the widths of columns A and B. You can do so by dragging the right boundaries of columns A and B

Entering/Editing Cell Entries

- Select cell
- Click in formula bar or press function key F2
- Enter/Edit cell content
  - Type in the desired information
  - Backspace key (removes character on left)
  - Delete key (removes character on right)
  - Highlight by dragging over characters to change, then type correction (will replace what is highlighted)
- Press Enter key

Step 7: Change the worksheet Name

- Right click on *Sheet1* tab and select Rename from the shortcut menu
- Enter a new name, for example *Gradesheet*



Step 8: Writing Formulas

- Points calculation for a course
  - Remember, the points are given by multiplying the credits with the numerical grade in the course
  - Thus for cell E9 which is suppose to show points for the course in cell A9, the formula will be =C9*D9. We select cell E9 and enter this formula in the formula bar and press Enter. Cell E9 should now show the result
  - Write similar formulas for cells E10 to E13

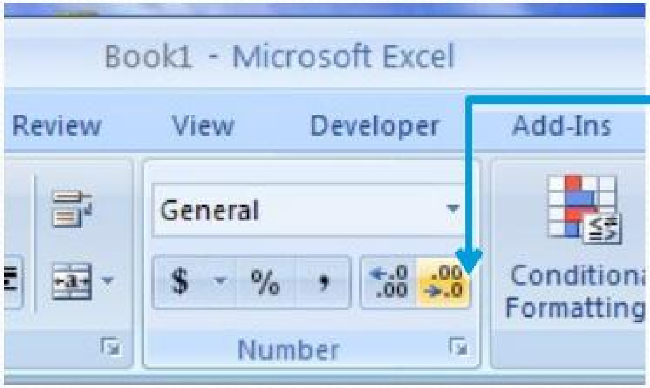
Step 8: Writing Formulas (Contd.)

- Formula for Total Credit Hours in cell C15
  - The total credit hours are given by adding credit hours from different courses
  - Thus for cell C15, the formula is =C9+C10+C11+C12+C13. You can also do the summation by using the built-in Excel function SUM and write the formula as =SUM(C9,C10,C11,C12,C13)

Important: Make sure you do not have a space preceding the equal sign while entering a formula

Step 8: Writing Formulas (Contd.)

- Formula for GPA in cell E15
  - The GPA is calculated by dividing the total points by the total credit hours
  - Thus for cell E15, the formula is =(E9+E10+E11+E12+E13)/C15
  - Note, the use of parenthesis to instruct Excel to add points first. Also note the use of already calculated total credit hours in C15
- You will see that Excel shows the result with many places after the decimal. Use the Decrease Decimal button to show only two places after the decimal

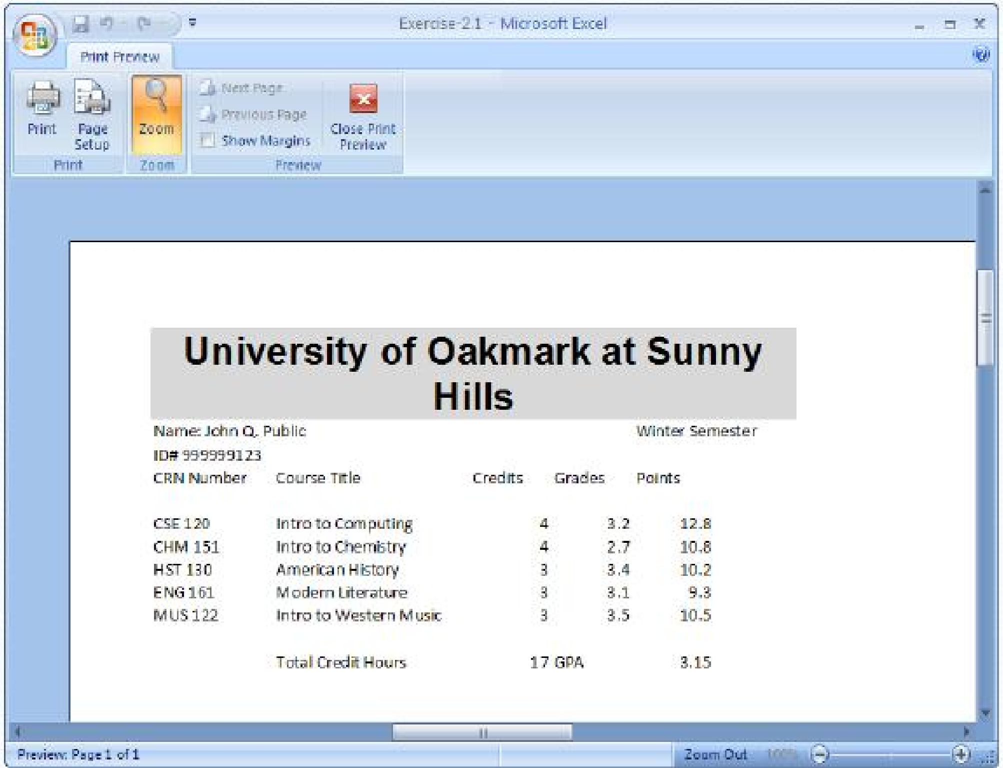


Step 9: Saving Your Worksheet

- Click on the File tab and select Save As command
- Select the Excel Workbook option
- In the ensuing dialog box, enter an appropriate name for your workbook and click Save
- You will notice the name you have given to your workbook now appears in the title bar at the top replacing the default name *Book1*

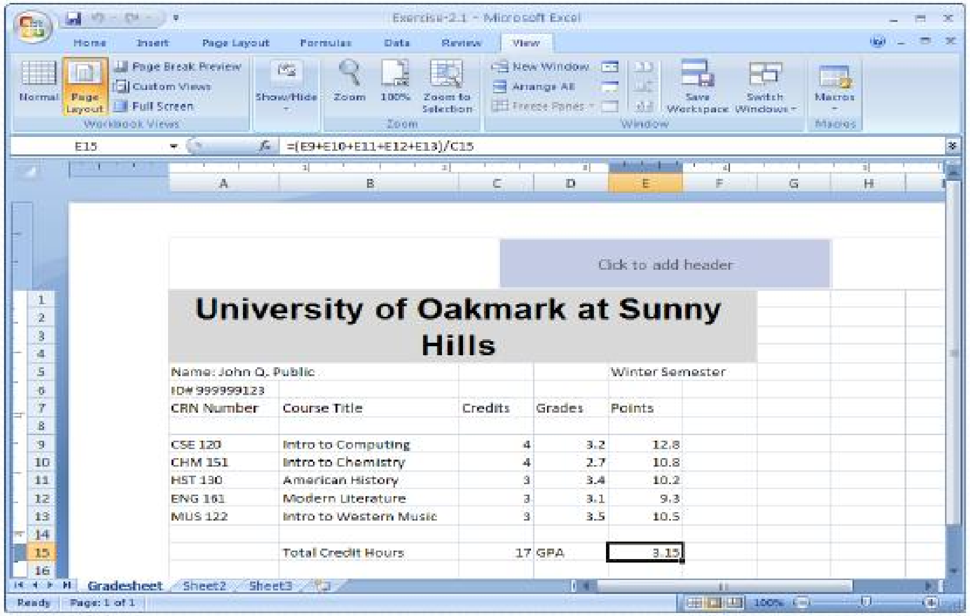
Printing a Worksheet

- Click the Office Button and select the Print command
- Select the print settings through the Print dialog box
- Use the Print Preview option to preview your sheet before printing



Printing a Worksheet

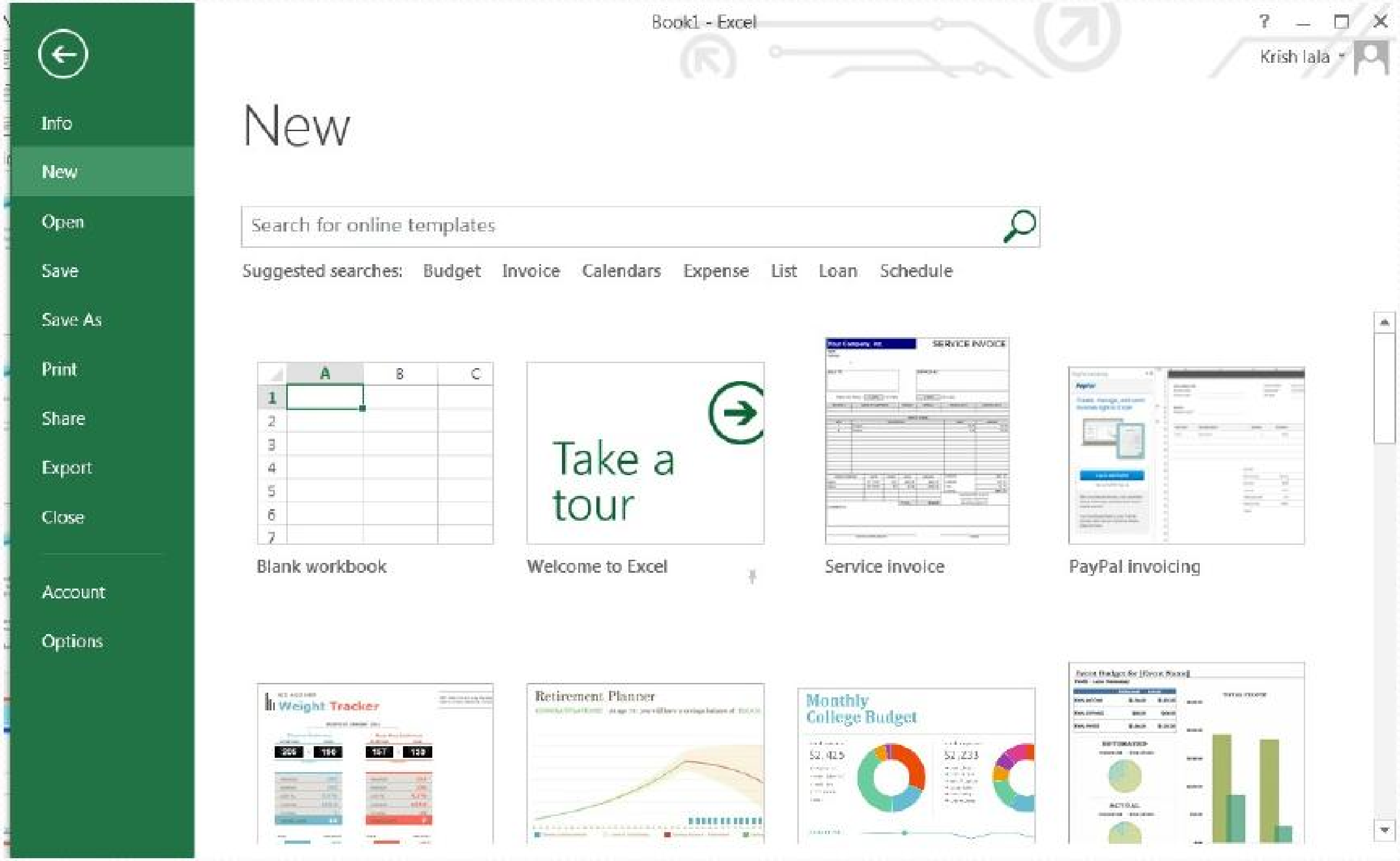
- You can also use the View tab for previewing and printing
  - Click on the View tab to make it active
  - Click the Page Layout button in the Workbook Views panel
  - Click the line *Click to add header* and type the desired header



Worksheet Templates

- A template is an Excel file that is already formatted, has formulas, and cells marked for data entry. You fill in your specific information and cell values to create a working sheet from it
- Excel comes with several templates. To open a template, select New from the File menu and then select the desired in the Backstage window
- Microsoft Office Online also provides many templates

Select/Search a Template

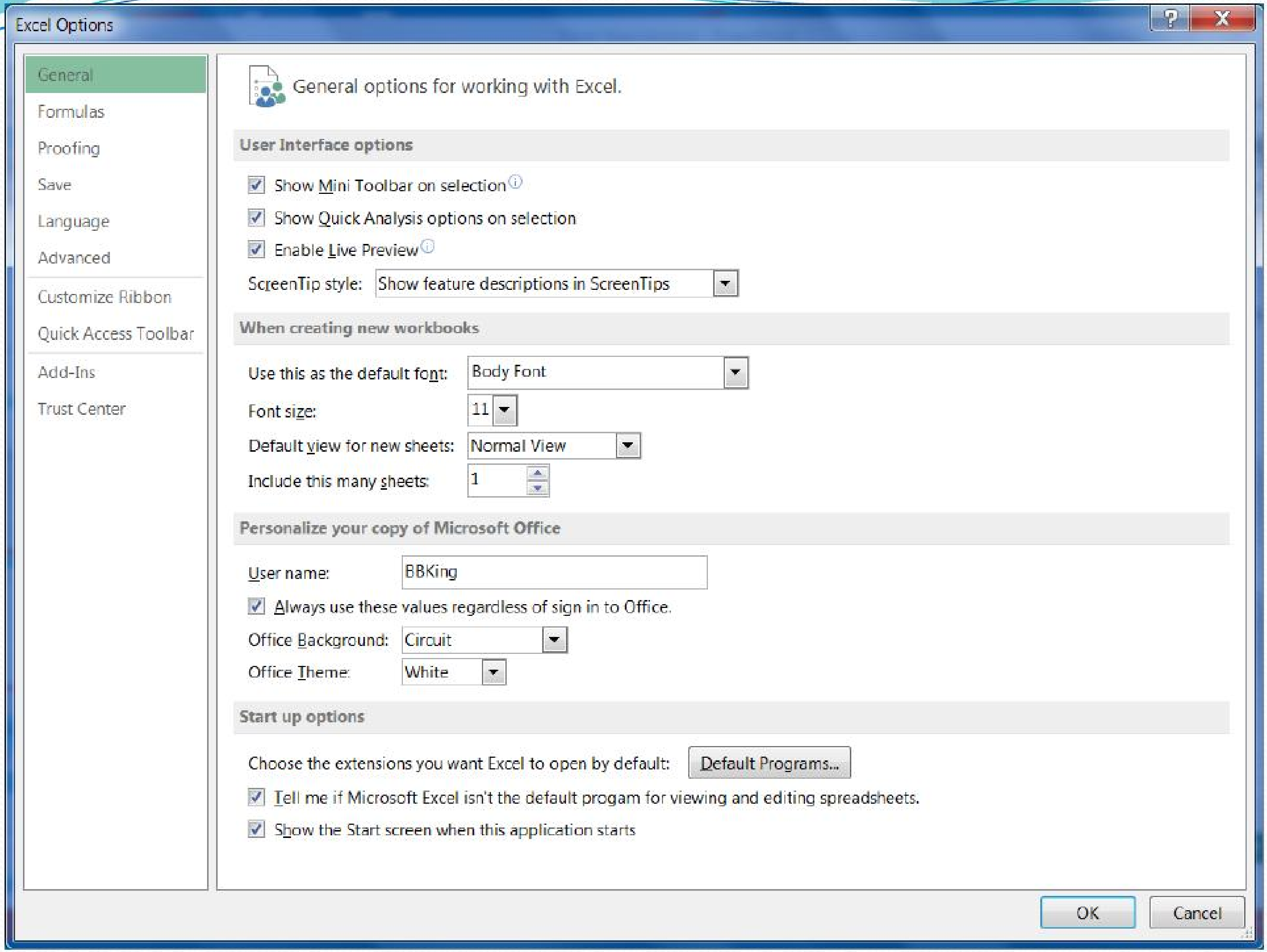


Excel 2013 File Formats

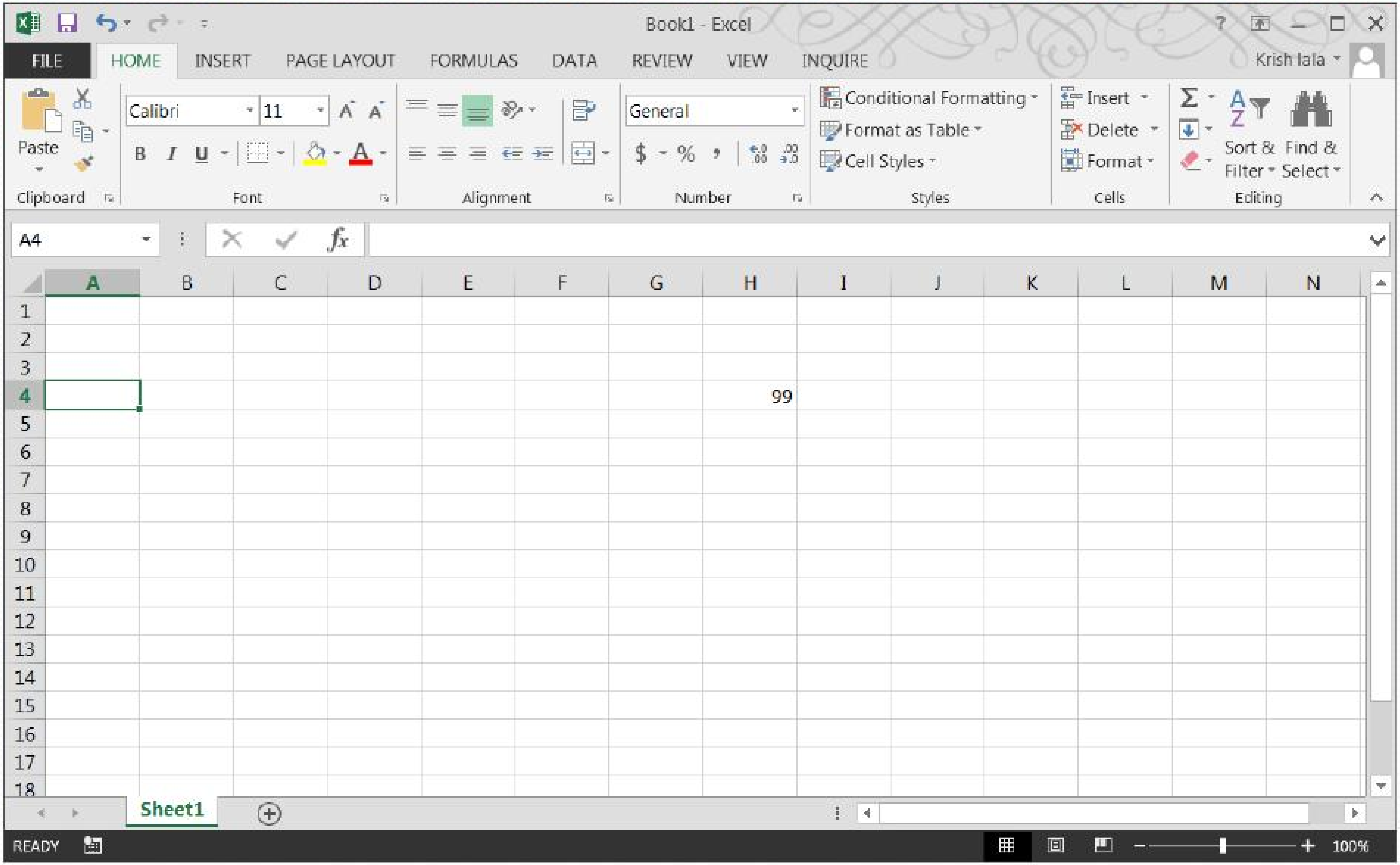
- Several formats are available
- Default format is .xlsx
- Saving in .xls (Excel 2003) format is advised when you are sharing your files with others
- Excel templates have .xltx format
- Workbooks with macros are saved with .xlsm extension

Customizing Excel Settings

- Click File > Option
- Several categories of options are available for customization
- Some examples of options:
  - Turn on/off the Mini Toolbar
  - Customize Excel window
  - Change the default font setting
  - Calculation mode

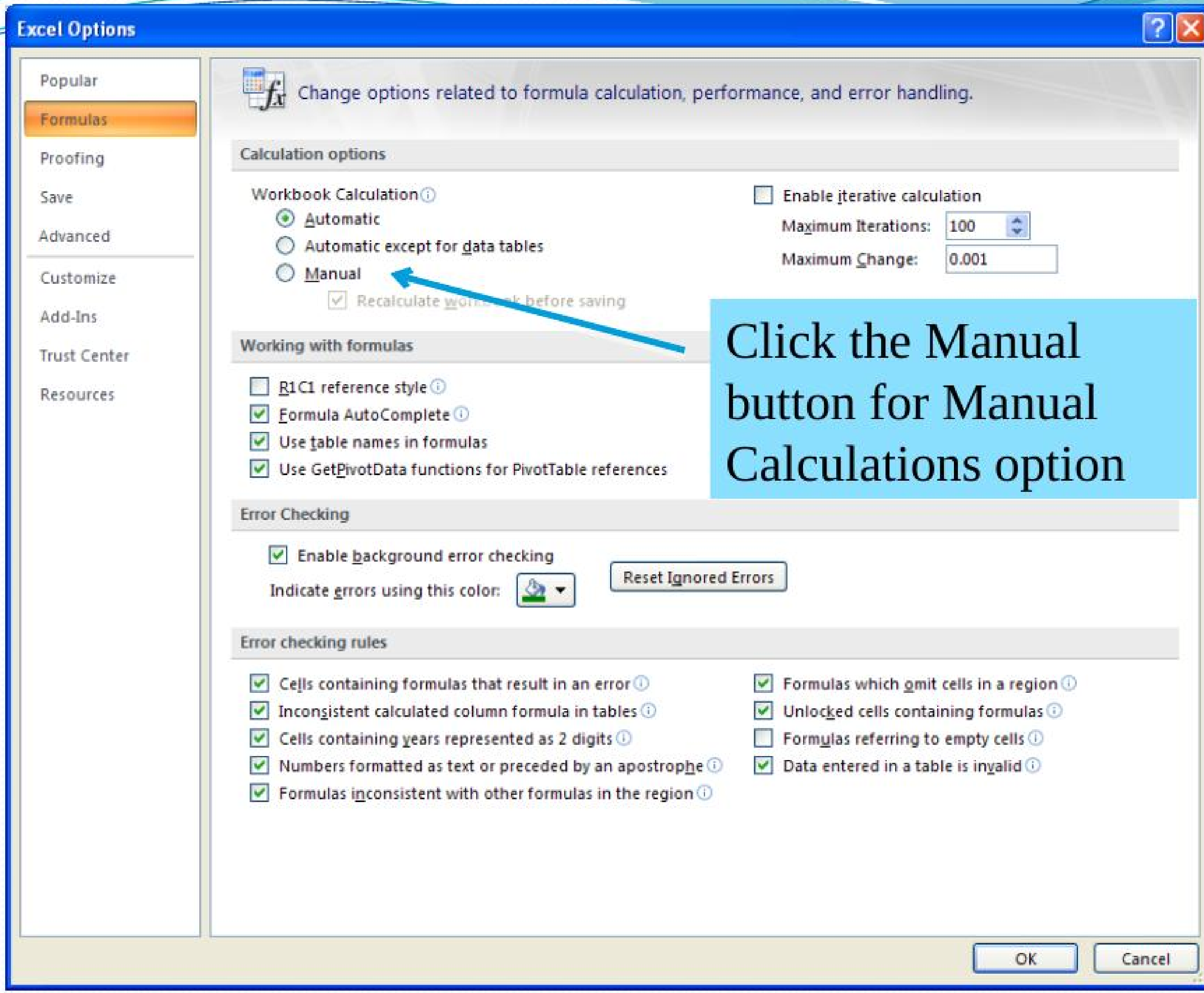

Excel Options
?
X
General
Formulas
Proofing
Save
Language
Advanced
Customize Ribbon
Quick Access Toolbar
Add-Ins
Trust Center
General options for working with Excel.
User Interface options
Show Mini Toolbar on selection
Show Quick Analysis options on selection
Enable Live Preview
ScreenTip style:
Show feature descriptions in ScreenTips
When creating new workbooks
Use this as the default font:
Body Font
Font size:
11
Default view for new sheets:
Normal View
Include this many sheets:
1
Personalize your copy of Microsoft Office
User name:
BBKing
Always use these values regardless of sign in to Office.
Office Background:
Circuit
Office Theme:
White
Start up options
Choose the extensions you want Excel to open by default:
Default Programs...
Tell me if Microsoft Excel isn't the default progam for viewing and editing spreadsheets.
Show the Start screen when this application starts
OK
Cancel

Excel Window with a Different Color Scheme



Calculation Modes

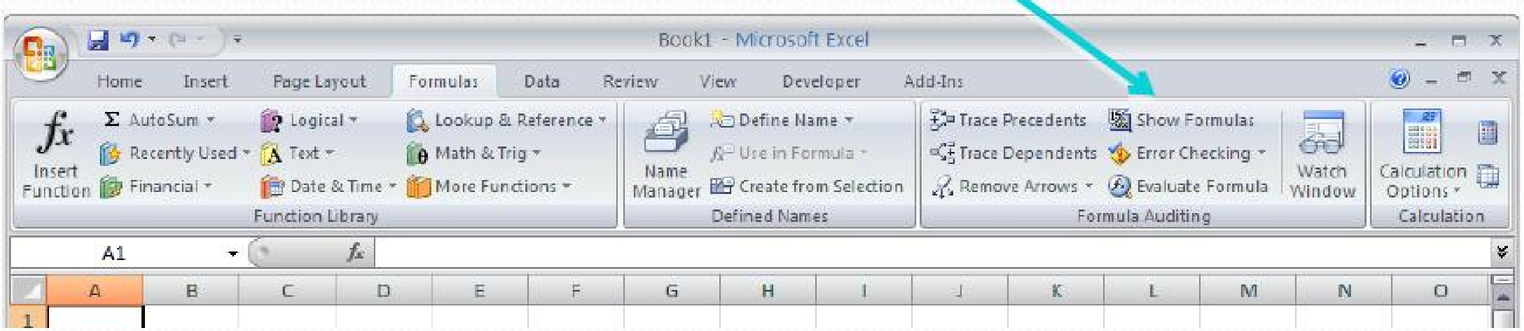
- Excel automatically updates the results of formulas as you make changes in cells referenced in formulas
- You can also set Excel to manual calculation mode. In this mode Excel updates the calculation results only after you press the function key F9
- You can do this in Excel Options window by selecting the Formula category of options



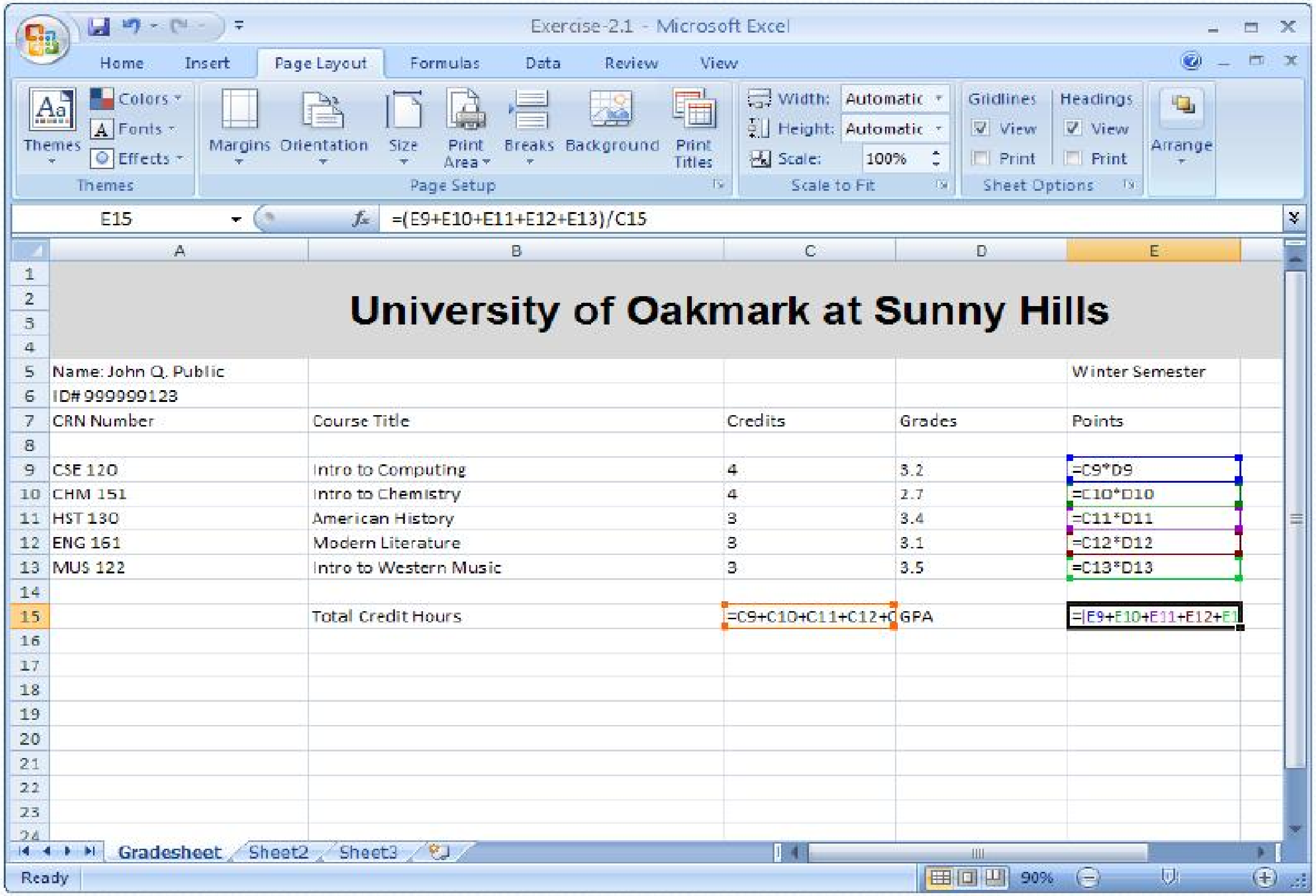
Click the Manual button for Manual Calculations option

Formula View

- In the Formula View, Excel shows cell formulas in place of showing the formula results
- The Formula View is good for sharing a worksheet to show how the calculations are being performed
- To select the Formula View, click on the Formulas tab, and then click on Show Formulas button in the Formula Auditing panel. You can do the same via Excel Options also

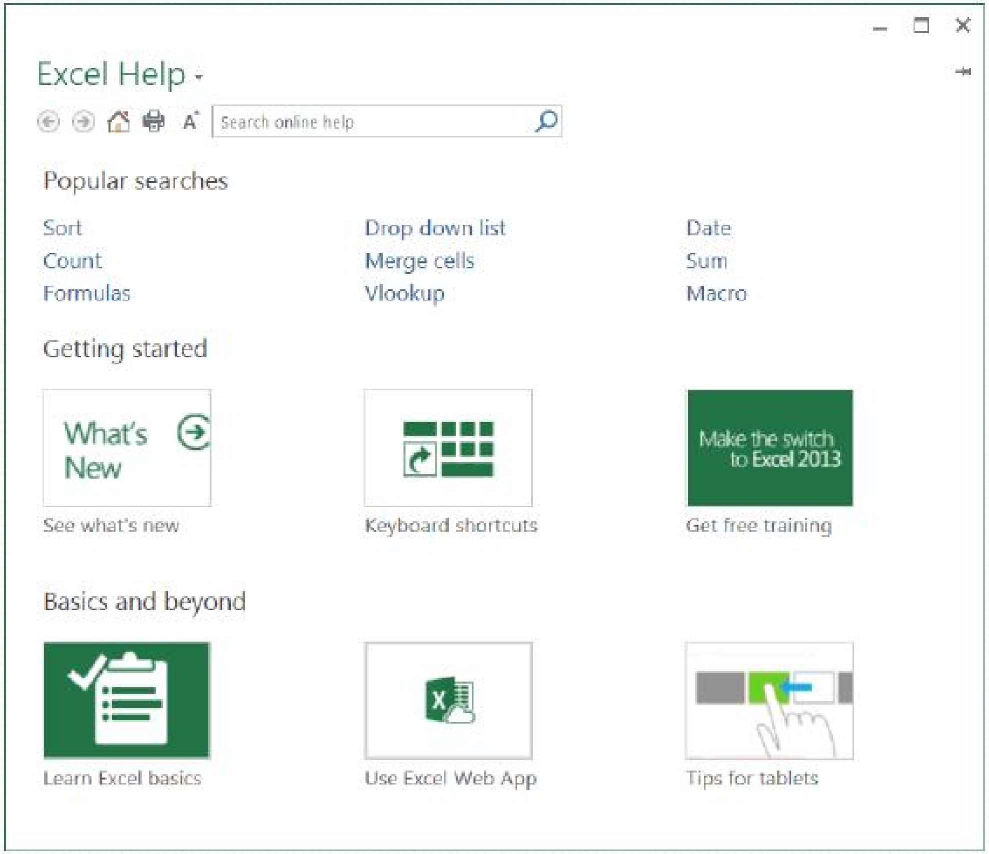


Formula View of the Completed Grade Calculation Worksheet



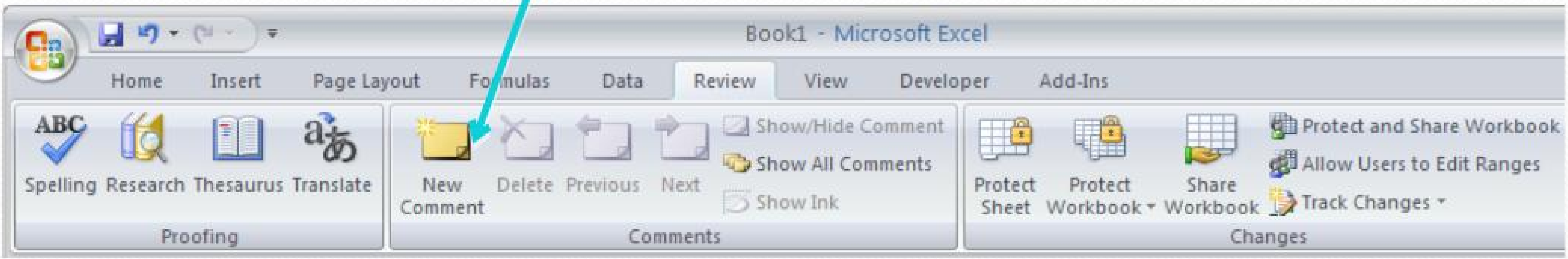
Seeking Help in Excel

- The question mark in the upper right corner of the Ribbon stands for Excel Help. You can also invoke Help by pressing the function key F1
- Excel responds by opening the Excel Help window where you can browse through help topics or do a search

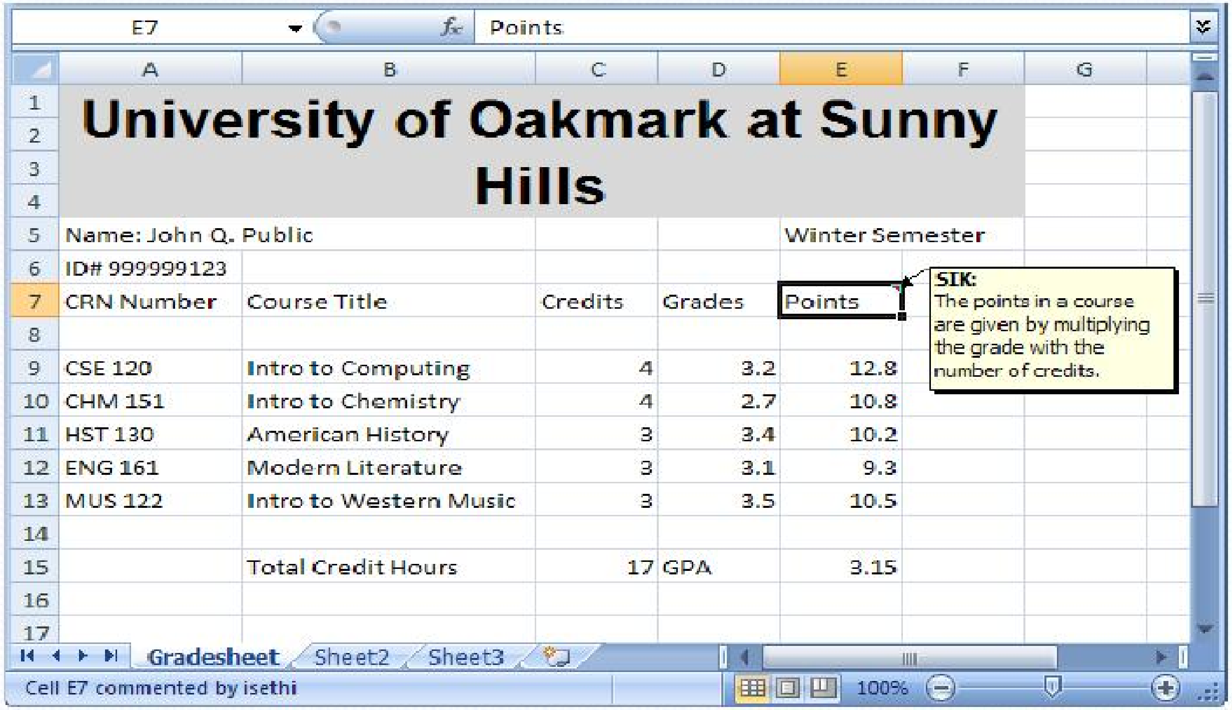


Adding Comments to Cells

- It is a good practice to add comments to cells with formulas for better understanding and sharing of worksheets
- To add comments to a cell
  - Select the cell
  - Make the Review tab active
  - Click the New Comments button in the Comments panel
  - Enter the comments and click on any other cell



An Example of a Cell with Comments



Thank You!